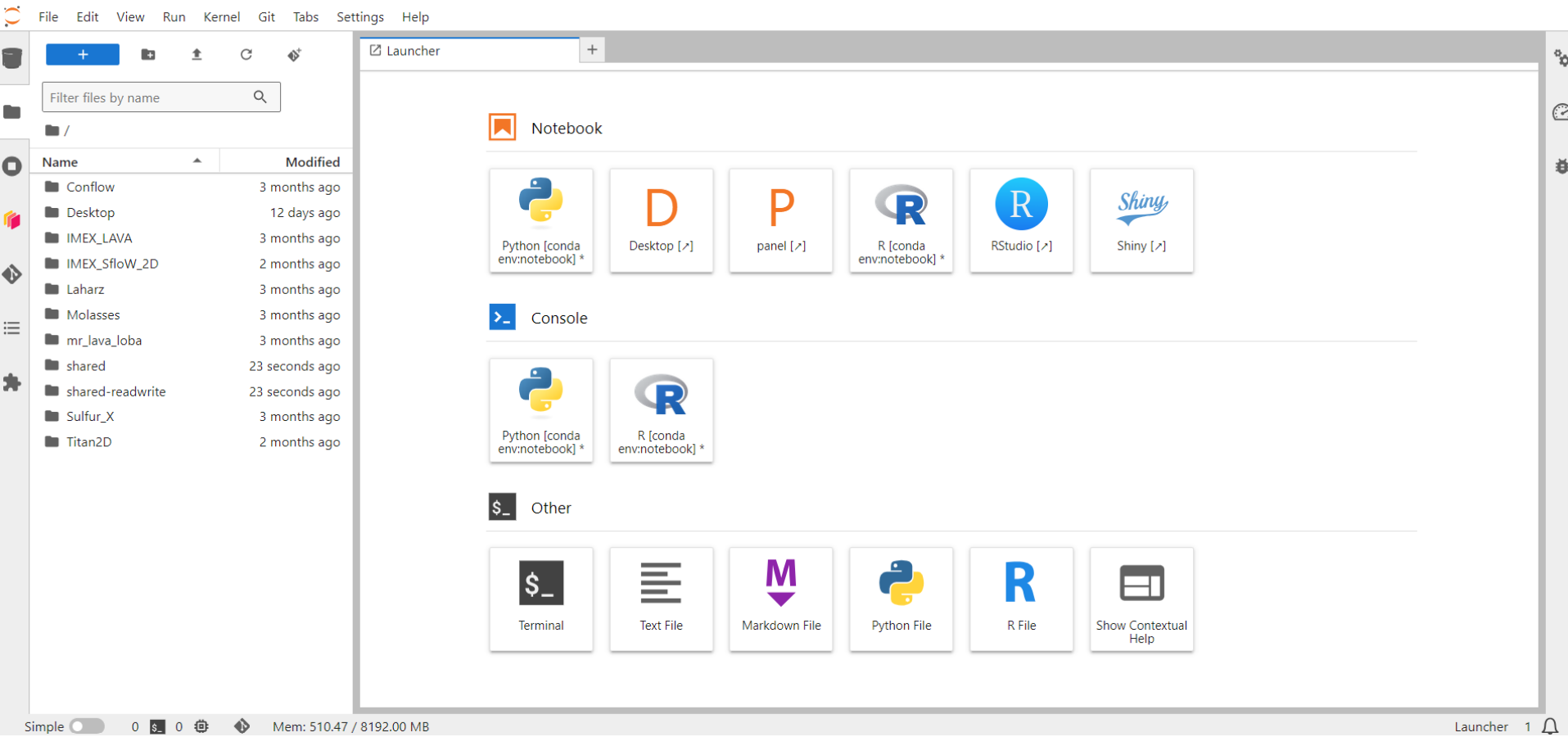


Intro to GIS

Exercise:

QGIS





- When you login to VICTOR, your screen will look something like this
- Your files are on the left panel
- Your “Launcher” has useful icons

Tip: If you lose your “Launcher” screen, click the “plus” icon at the top left for “New Launcher”

File Edit View Run Kernel Git Tabs Settings Help

Filter files by name

/ shared /

Name	Modified
CSAV	11 days ago
DEMs	14 days ago
Iceland_Nov2023	7 months ago
Libraries	14 days ago
Models	14 days ago
Scripts	13 days ago
VICTOR-short-courses	2 months ago
Workflows	29 days ago
Workshop	9 months ago
README.md	last year

shared

Notebook

Python [conda env:notebook] *

Desktop [↗]

panel [↗]

R [conda env:notebook] *

RStudio [↗]

Shiny [↗]

Console

Python [conda env:notebook] *

R [conda env:notebook] *

Other

Terminal

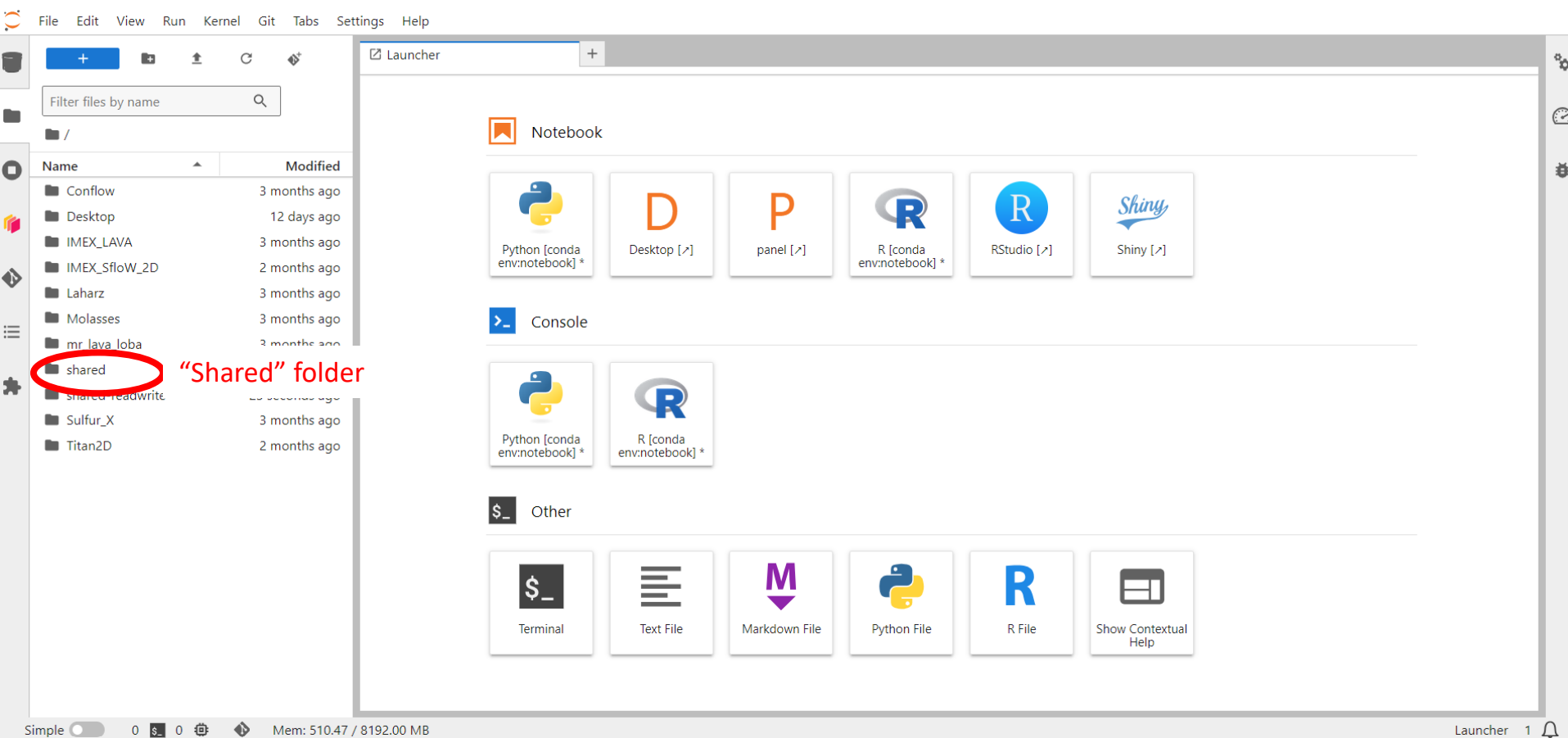
Text File

Markdown File

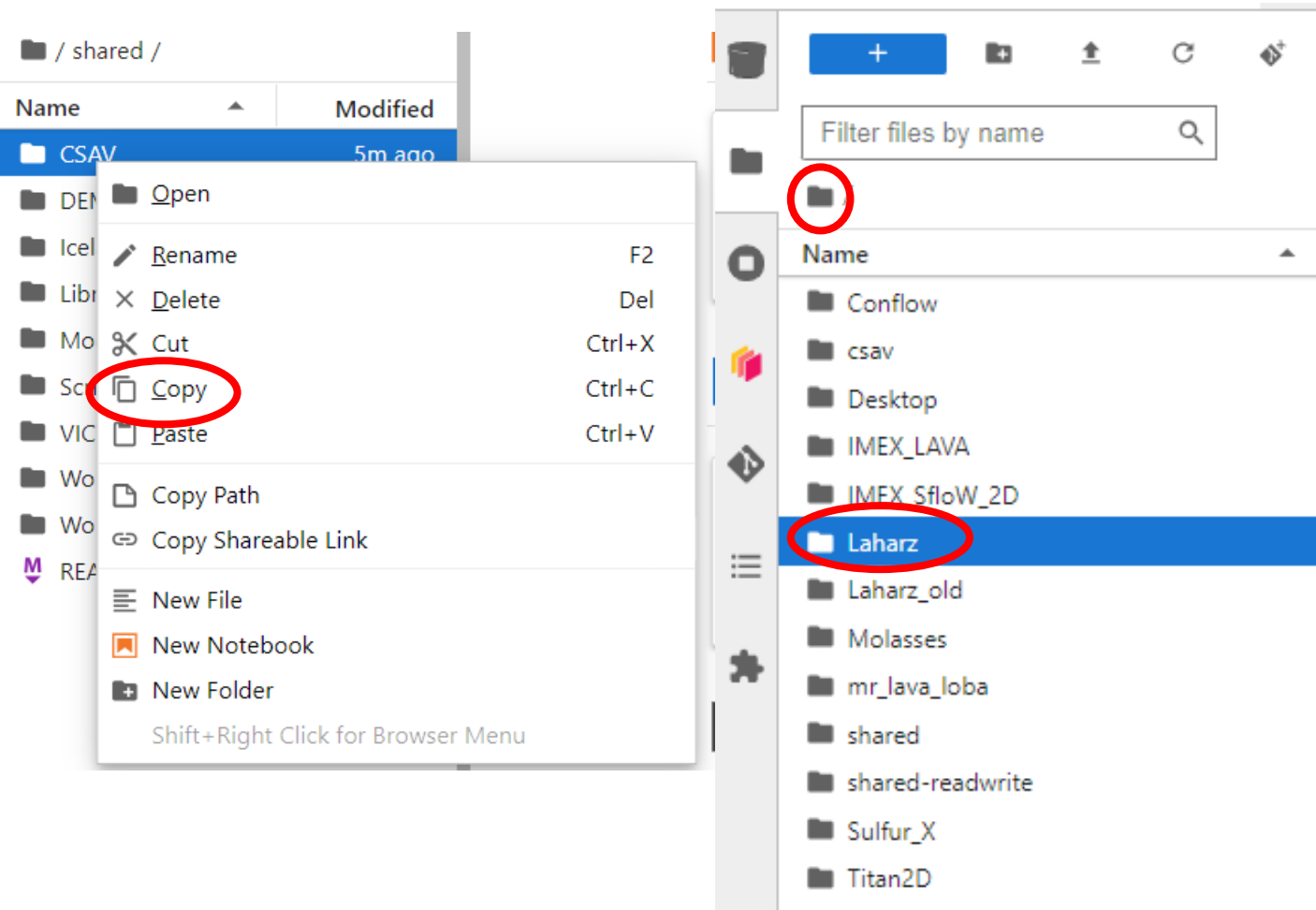
Terminal 3

```
+RANLIB=/srv/conda/envs/notebook/bin/x86_64-conda_cos6-linux-gnu-ranlib
+READELF=/srv/conda/envs/notebook/bin/x86_64-conda_cos6-linux-gnu-readelf
+SIZE=/srv/conda/envs/notebook/bin/x86_64-conda_cos6-linux-gnu-size
+STRINGS=/srv/conda/envs/notebook/bin/x86_64-conda_cos6-linux-gnu-strings
+STRIP=/srv/conda/envs/notebook/bin/x86_64-conda_cos6-linux-gnu-strip
INFO: activate-gcc_linux-64.sh made the following environmental changes:
+CC=/srv/conda/envs/notebook/bin/x86_64-conda_cos6-linux-gnu-cc
+CFLAGS=-march=nocona -mtune=haswell -ftree-vectorize -fPIC -fstack-protector-strong -fno-p
+CPPFLAGS=-D_FORTIFY_SOURCE=2 -O2
+CPP=/srv/conda/envs/notebook/bin/x86_64-conda_cos6-linux-gnu-cpp
+DEBUG_CFLAGS=-march=nocona -mtune=haswell -ftree-vectorize -fPIC -fstack-protector-strong
var-tracking-assignments
+GCC_AR=/srv/conda/envs/notebook/bin/x86_64-conda_cos6-linux-gnu-gcc-ar
+GCC_NM=/srv/conda/envs/notebook/bin/x86_64-conda_cos6-linux-gnu-gcc-nm
+GCC_RANLIB=/srv/conda/envs/notebook/bin/x86_64-conda_cos6-linux-gnu-gcc-ranlib
+GCC=/srv/conda/envs/notebook/bin/x86_64-conda_cos6-linux-gnu-gcc
+LDFLAGS=-Wl,-O2,--sort-common,--as-needed,-z,relro,-z,now
(notebook) jovyan@jupyter-sogburn:~$ victor setup
Hello, and welcome to the VICTOR platform.
We have a huge variety of models and tools for use.
This program is meant to make it simple to set up workflows and associated files.
To set up, please choose the number next to the model listed below.
1: CONFLOW
2: DISGAS
3: HAZMAP
4: HYSPLIT
5: IMEX (Lava)
6: IMEX_Sflow2D_V2
7: Laharz
8: LAVA2D
9: MR LAVA LOBA
10: MOLASSES
11: PYFLOWGO/DOWNFLOWGO
12: Scoops3D
13: SULFUR_X
14: TEPHRA2
15: TITAN2D
16: TWODEE
Model: 7
```

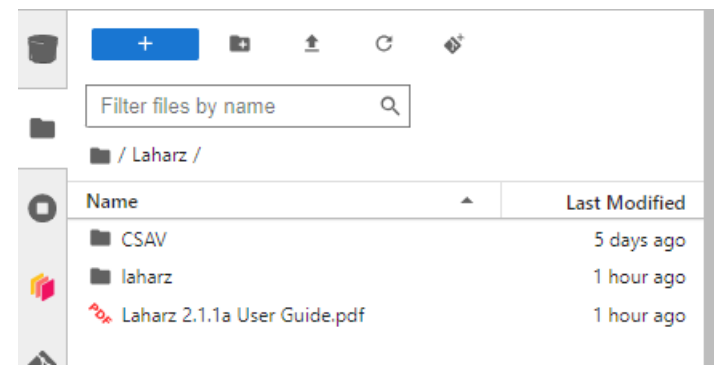
- We want to add the Laharz folder to your home workspace
- Click the terminal
- Type: victor setup
- Hit enter on your keyboard
- Type: 7
- Hit enter
- Close the terminal



- Double click the “shared” folder



- Right click on the “CSAV” folder
- Click “Copy”
- Navigate to your home folder (click the folder icon)
- Navigate to the “Laharz” folder
- Paste the “CSAV” folder there



Filter files by name

/

Name	Modified
Conflow	3 months ago
Desktop	12 days ago
IMEX_LAVA	3 months ago
IMEX_SfIoW_2D	2 months ago
Laharz	3 months ago
Molasses	3 months ago
mr_lava_loba	3 months ago
shared	23 seconds ago
shared-readwrite	23 seconds ago
Sulfur_X	3 months ago
Titan2D	2 months ago

Launcher

Notebook

Click on the "Desktop" icon

Python [conda env:notebook] *

Desktop [↗]

panel [↗]

R [conda env:notebook] *

RStudio [↗]

Shiny [↗]

Console

Python [conda env:notebook] *

R [conda env:notebook] *

Other

Terminal

Text File

Markdown File

Python File

R File

Show Contextual Help

Applications :

Wed 5 Jun, 16:20 Default user



File System



Home



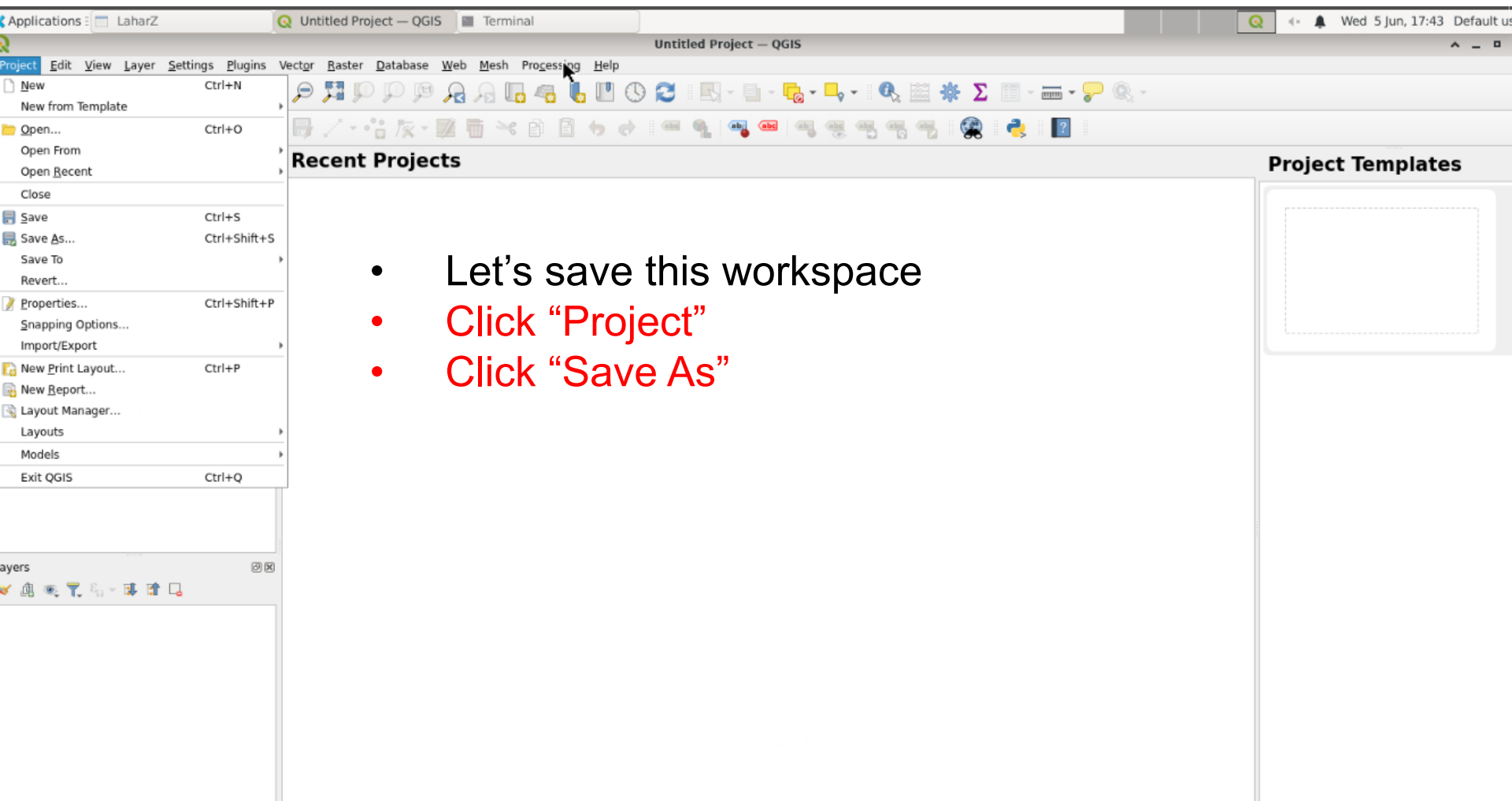
QGIS

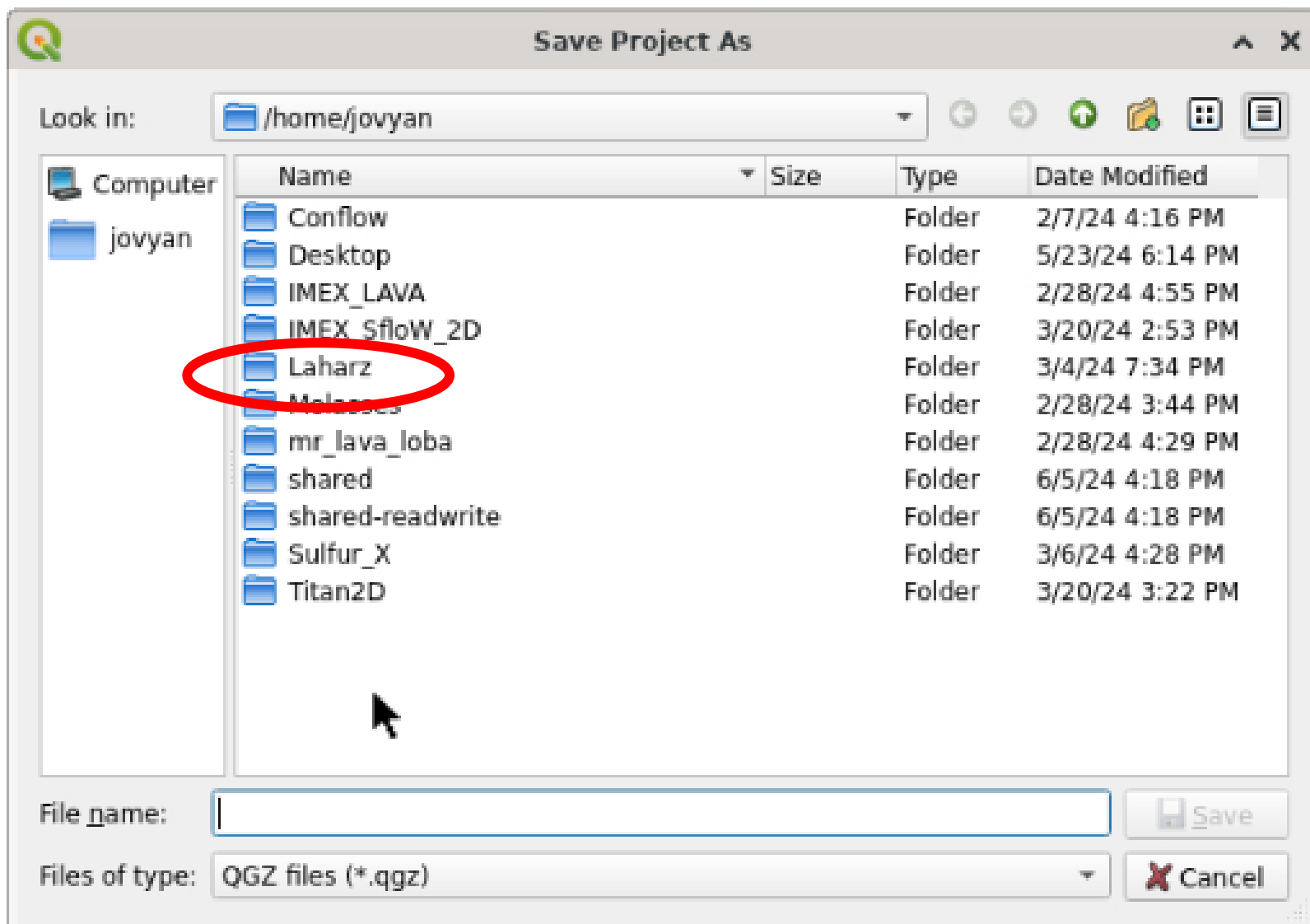


Paraview

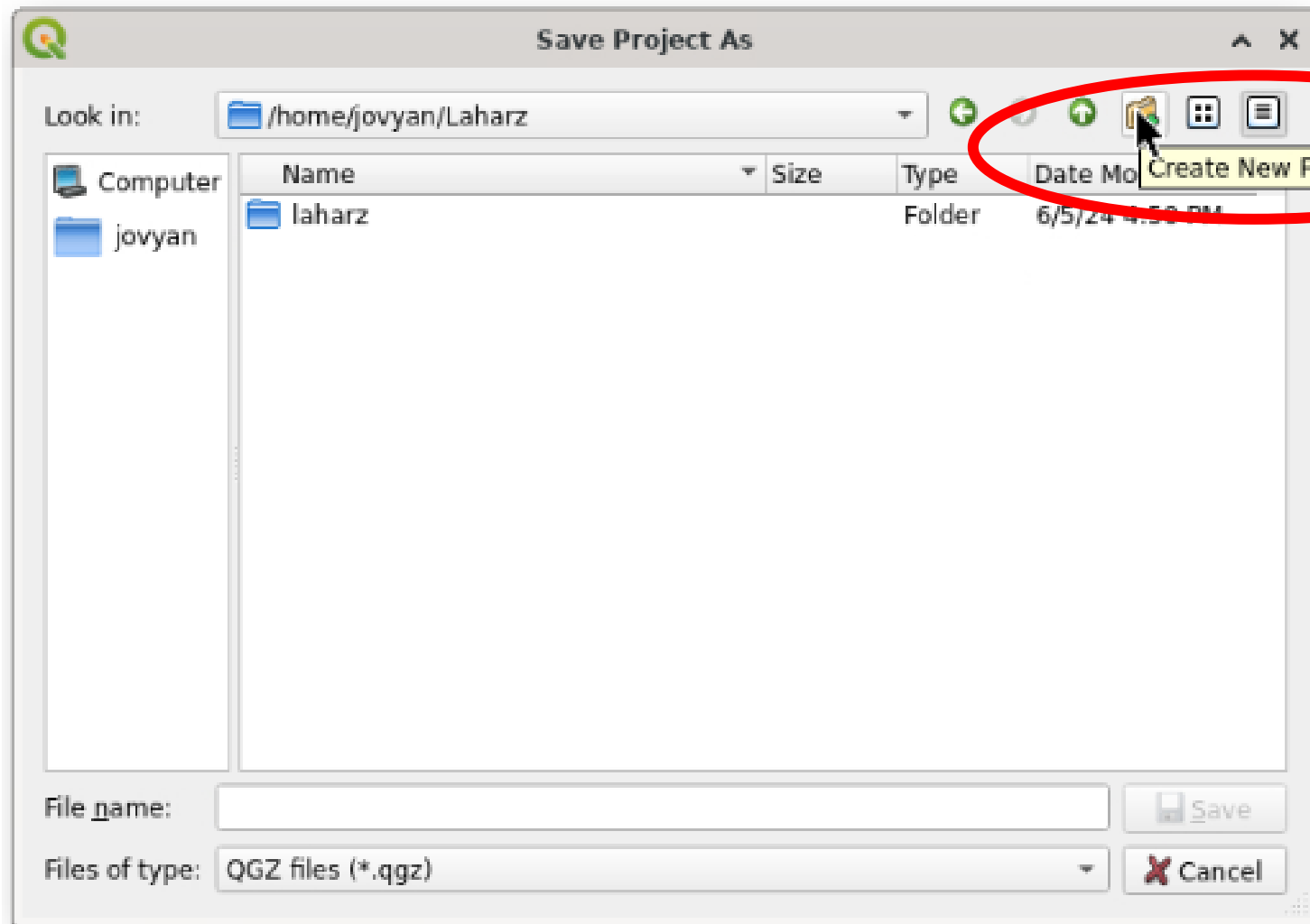
Click on QGIS, if you get an error message, choose "launch anyway"



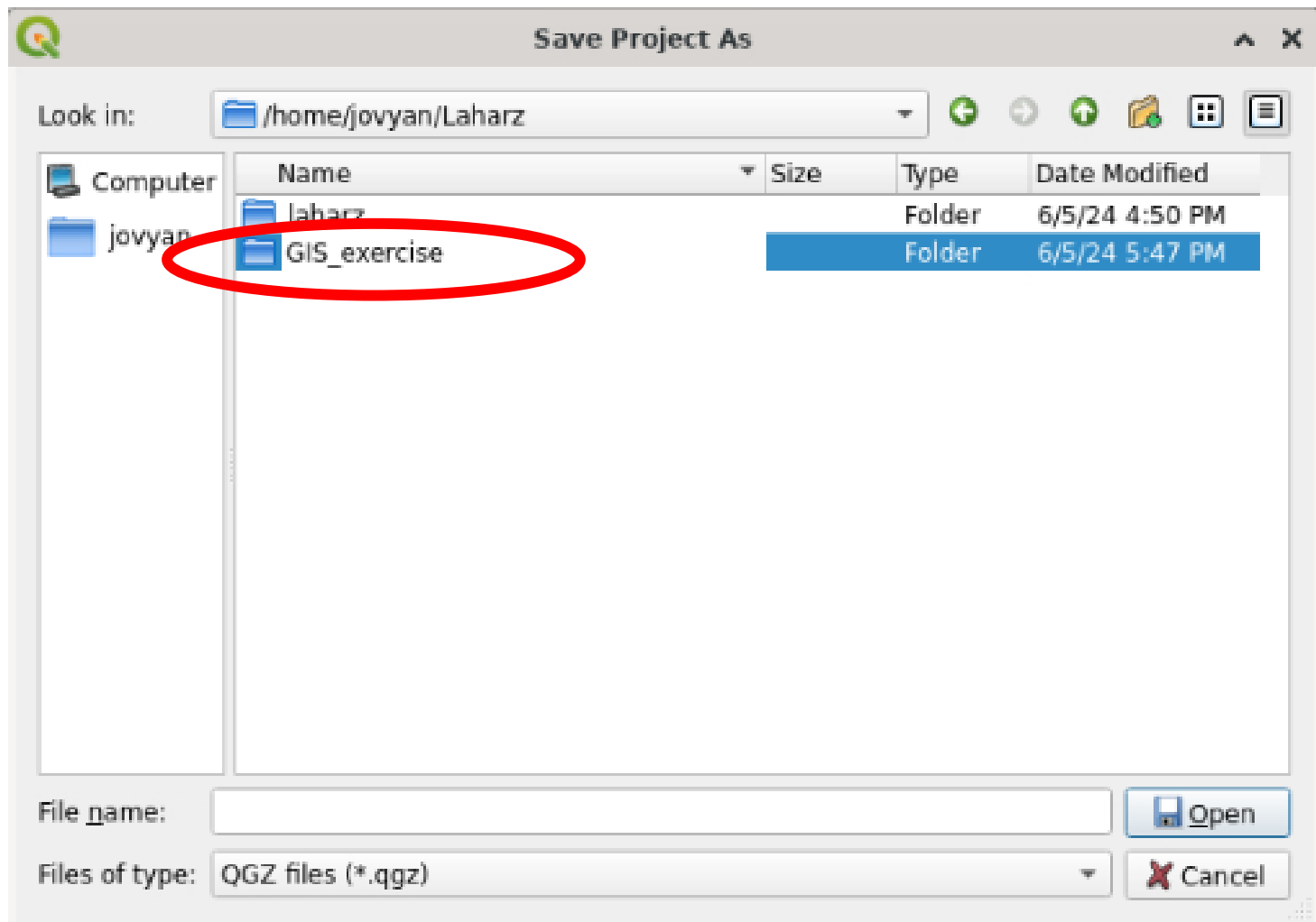




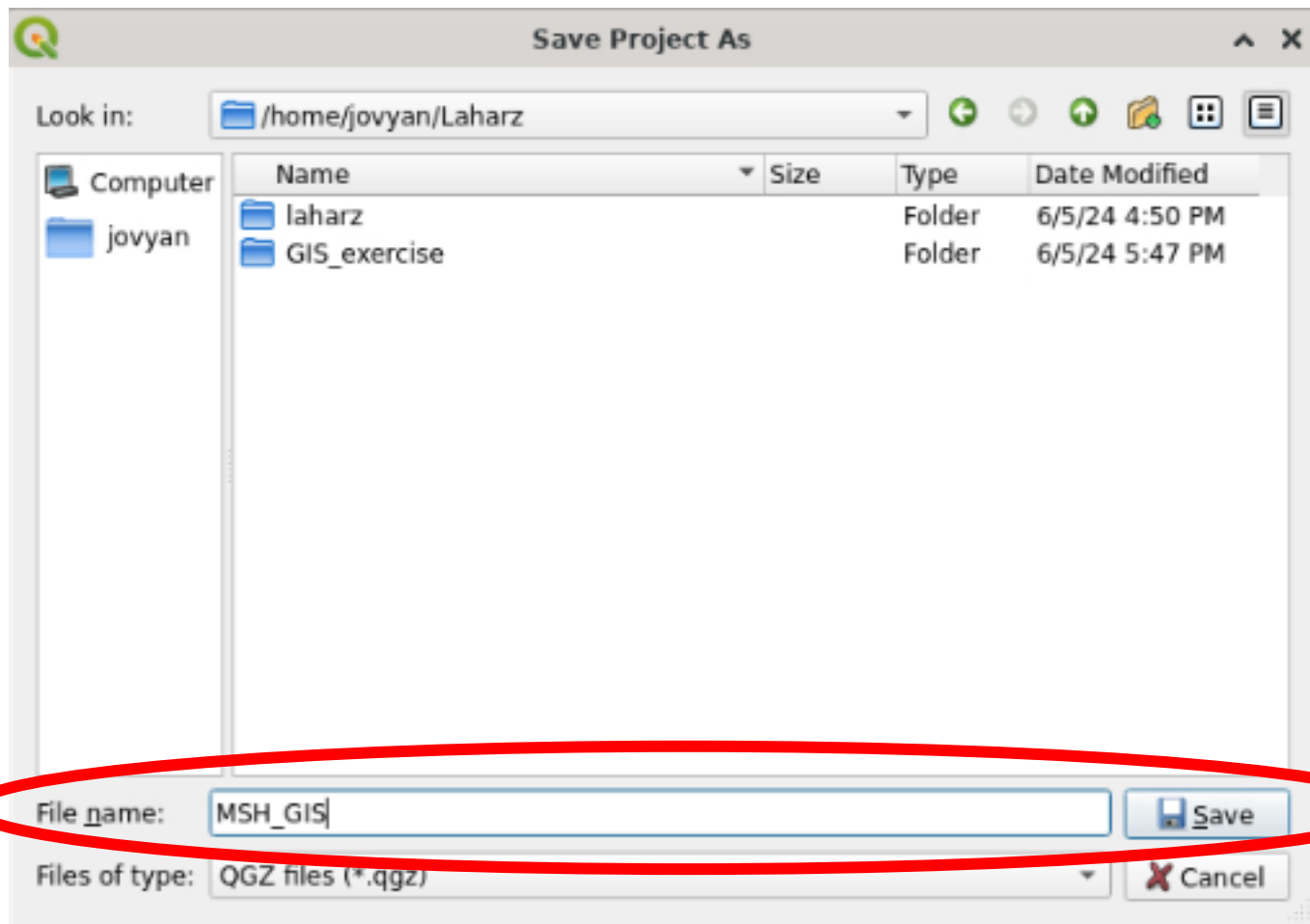
- Navigate to the “Laharz” folder



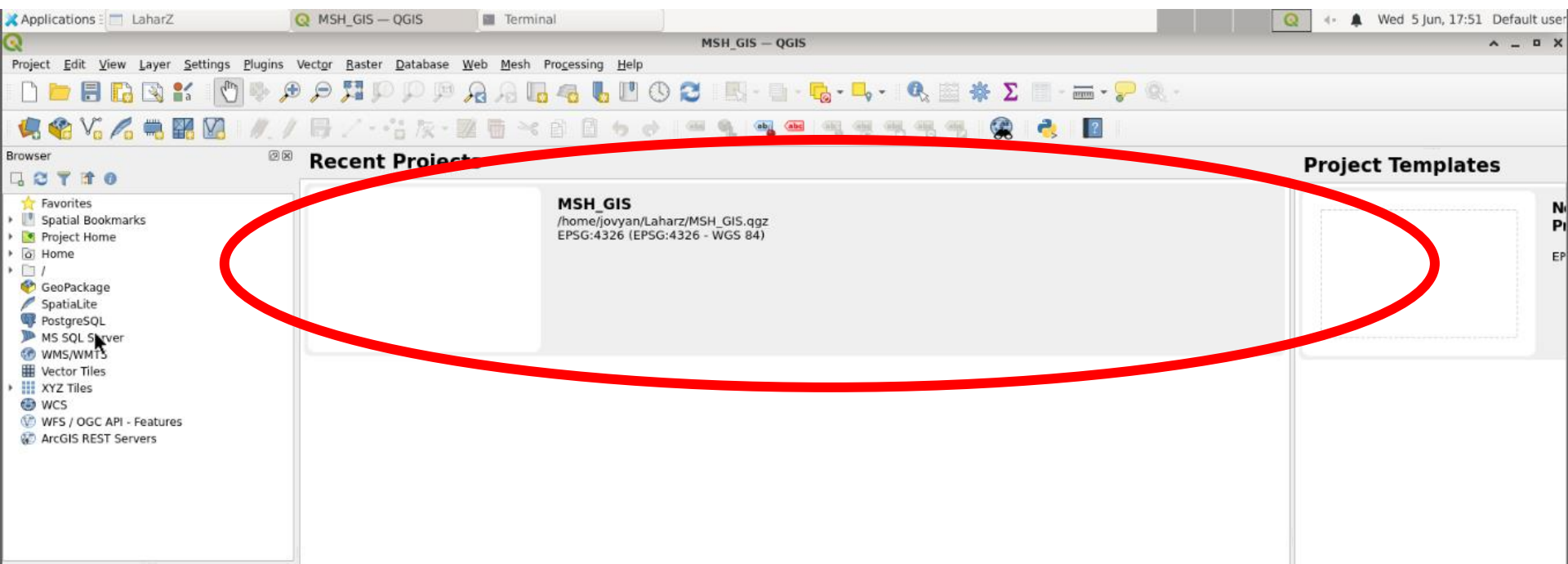
- Click “Create New Folder”



- Name the new folder “GIS_exercise” or something similar



- In the “File name” box, type the name of your project
- I named this “MSH_GIS”
- Click “Save”



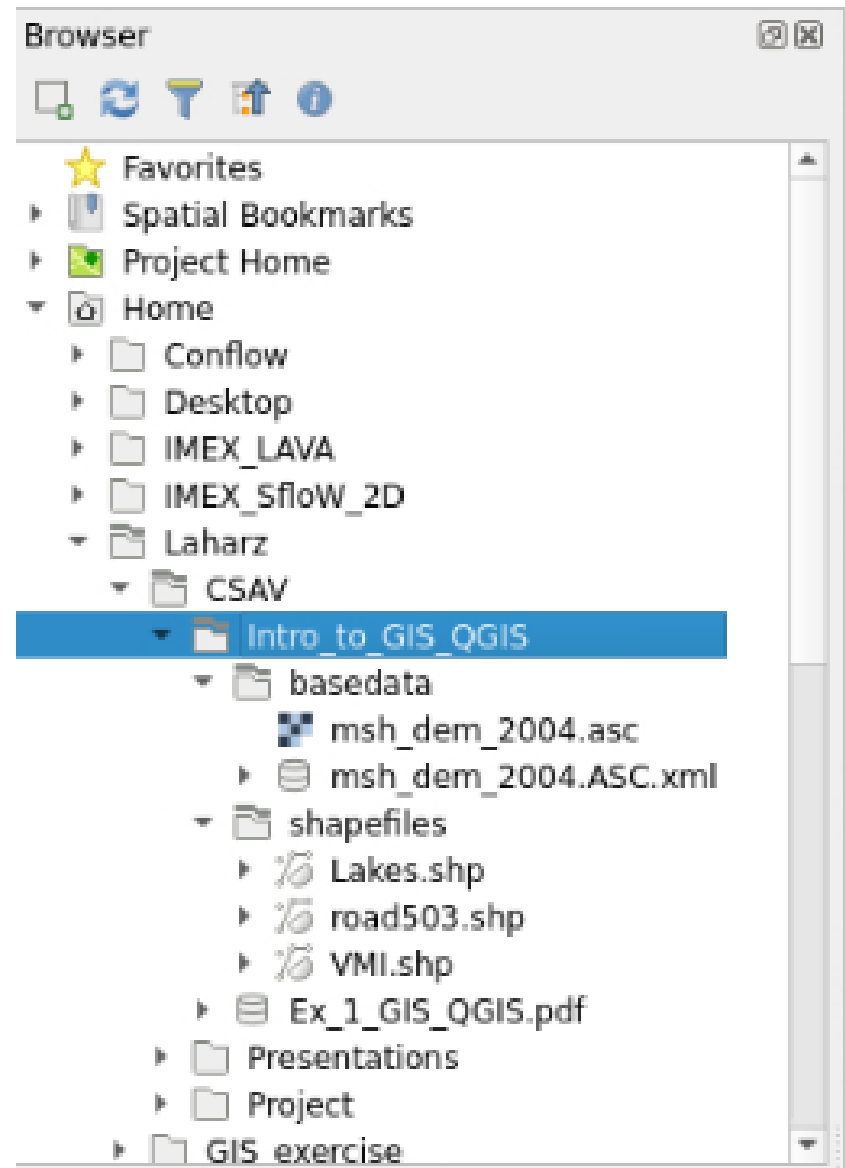
- Click on the new project

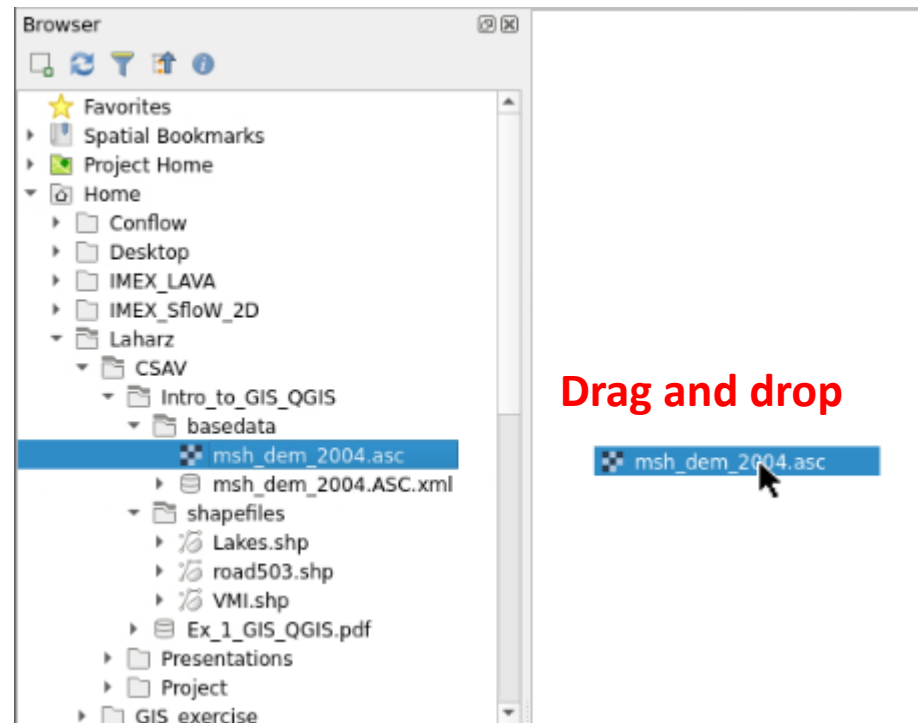
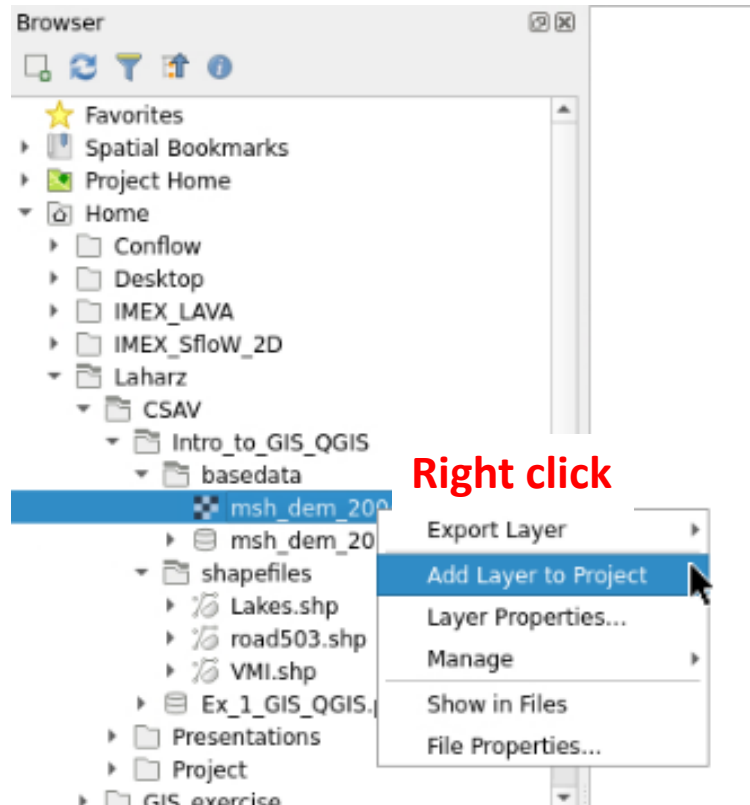
Exercise Goals

- **Add Instrument data set and background information for context**
- View Instrument attribute table
- Select instruments by drawing rectangle
- Select instruments by their attributes
- Assign colors to instrument type
- Add a legend to describe the meaning of colors

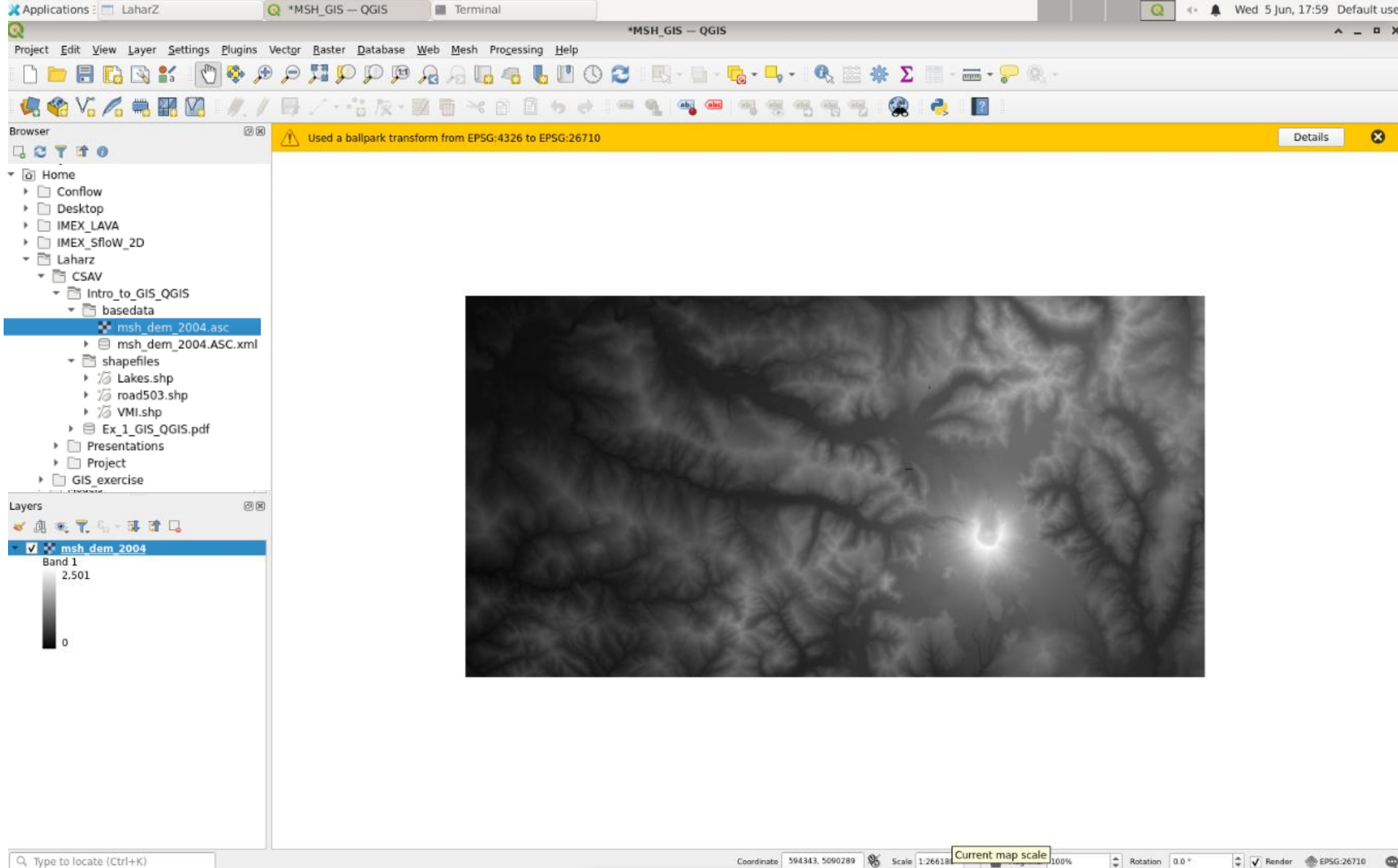
Navigate to files

- In the “Browser” panel of QGIS, navigate to the Laharz/CSAV/Ex_1_Intro_to_GIS_QGIS folder
- In this folder are two folders:
 - Basedata with the Mt. St. Helens 2004 DEM
 - Shapefiles with lake, road, and the volcano monitoring instrument (VMI) layers

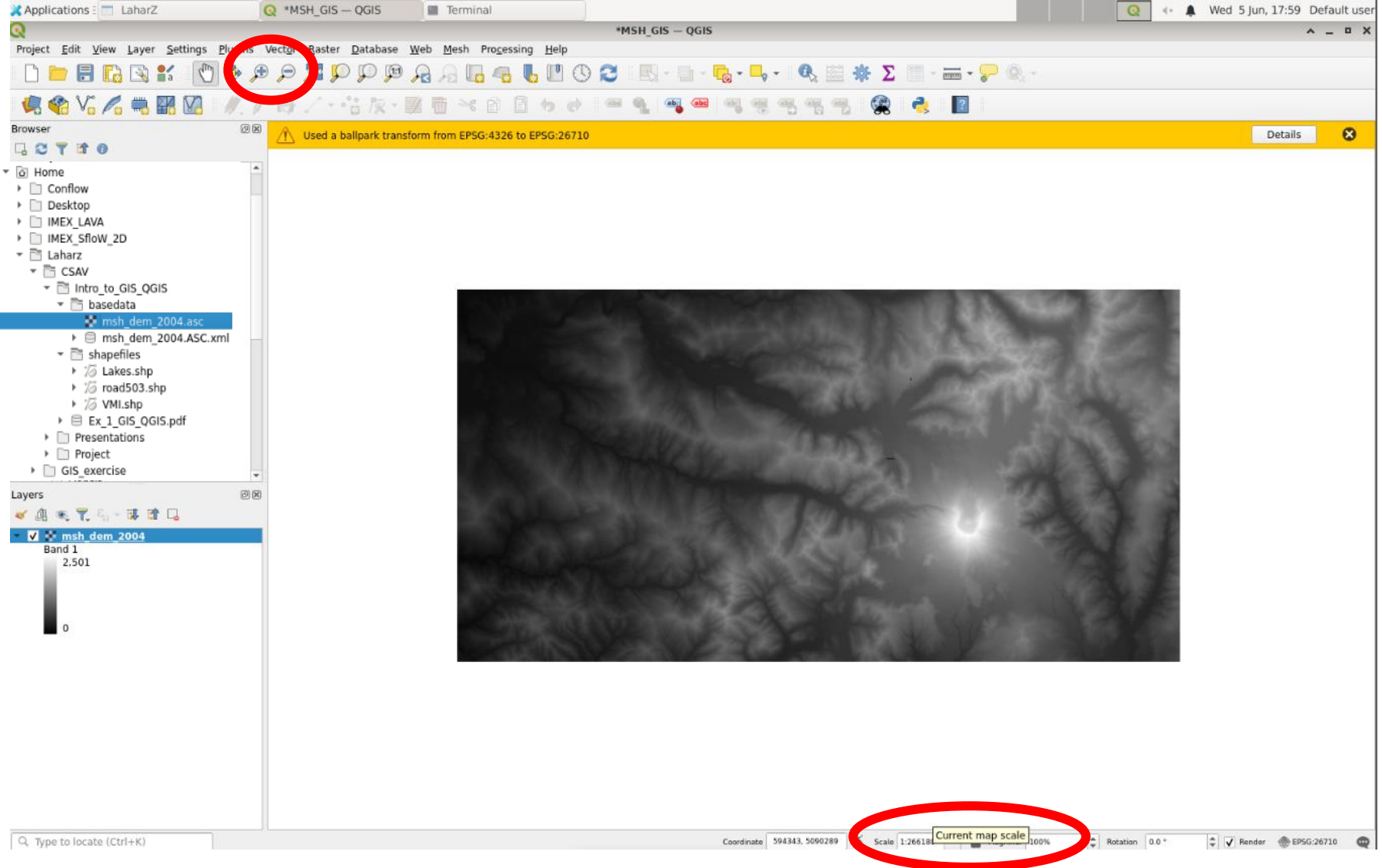




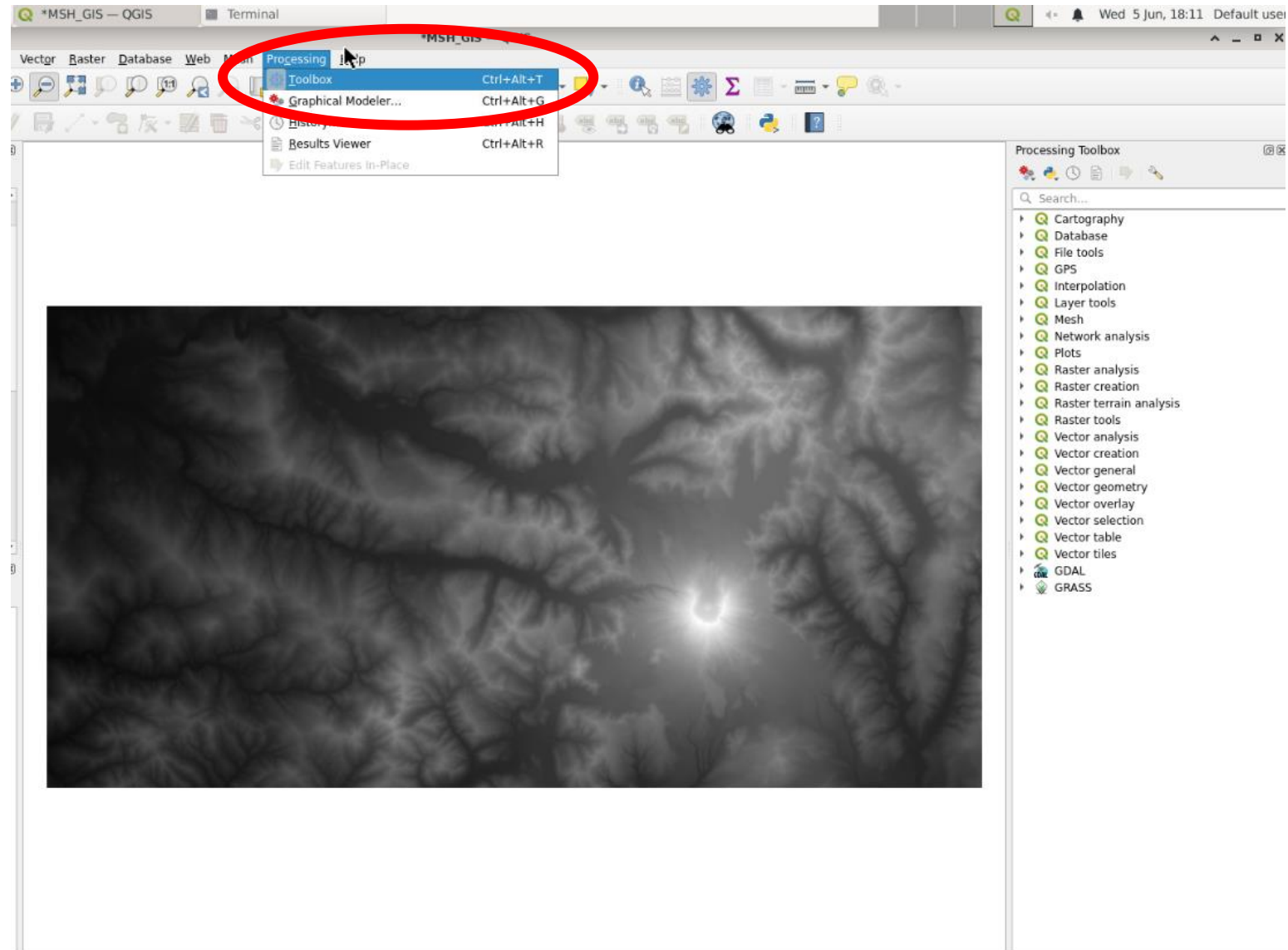
- Two ways to add layers:
 - Right click on the “msh_dem_2004.asc” file and choose “Add Layer to Project”
 - Drag the “msh_dem_2004.asc” layer and drop onto the workspace



- You should see the MSH DEM on the screen

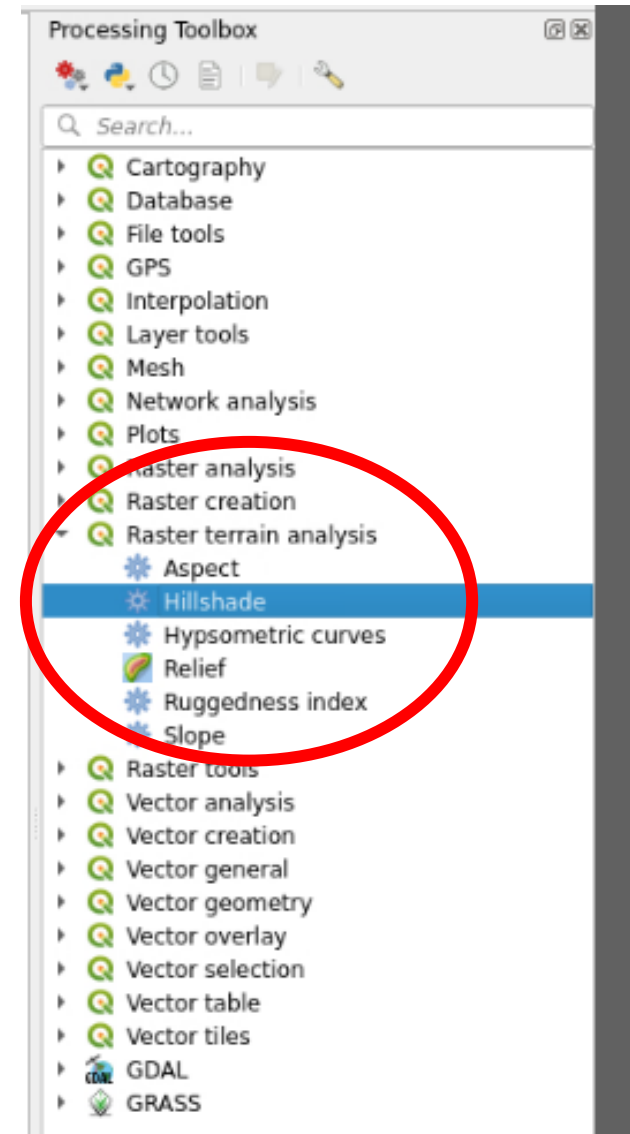


- You can zoom in and out 3 ways:
 - Use the mouse wheel
 - Click the magnifying glass icon, then click on the map
 - Edit the map scale manually at the bottom of the screen. Try 1:90000



- Open the Processing Toolbox

- Let's make a hillshade
 - A hillshade (or shaded-relief model) is a rendition of shadows across a DEM that helps our brains interpret what we see in DEMs.
 - Click the “Raster terrain analysis” menu
 - Double click the “Hillshade” tool
 - You can also use the “Search” field to search for tools!



Let's make a hillshade

- This tool let's you model a “sun” and the “shadows” it would produce on your DEM
- Use the default parameters
 - Z factor is vertical exaggeration (if your z units differ from your x and y units)
 - The azimuth is where the sun is relative to your map. The human brain expects the shadows to be modeled with the sun coming from the upper left (300°)
 - The vertical angle is how high in the sky the modeled sun is. We use the default 40°

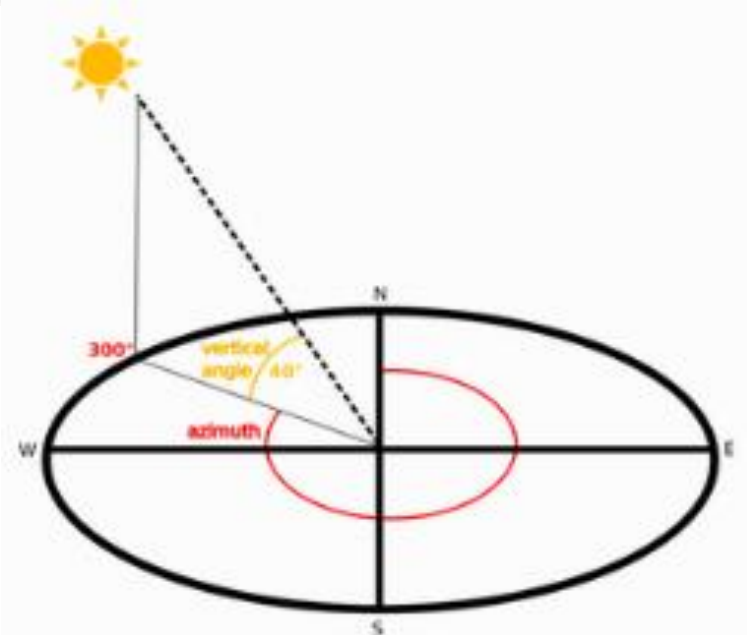
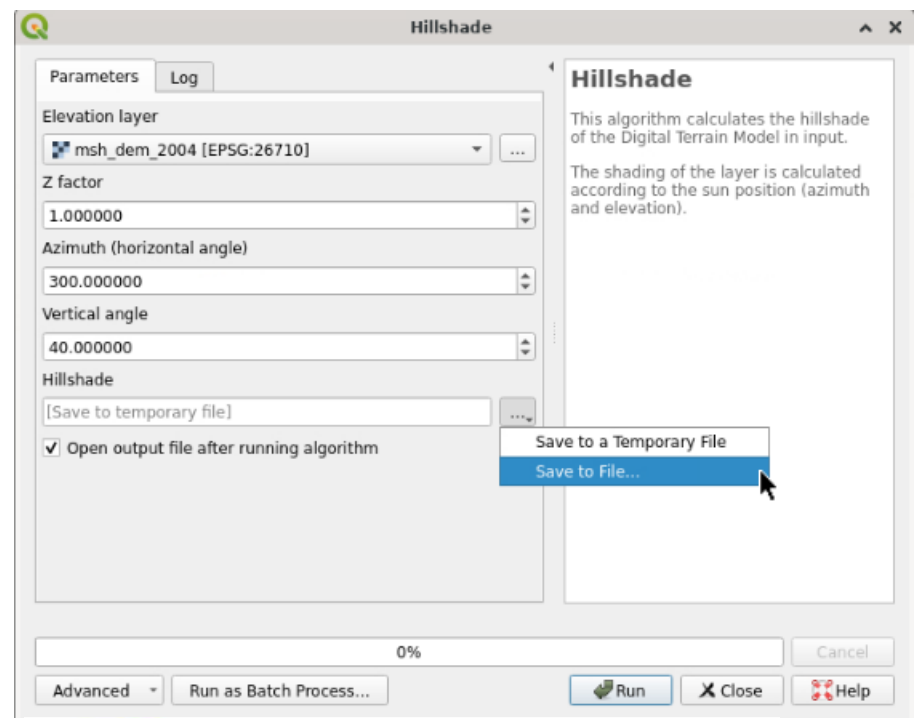
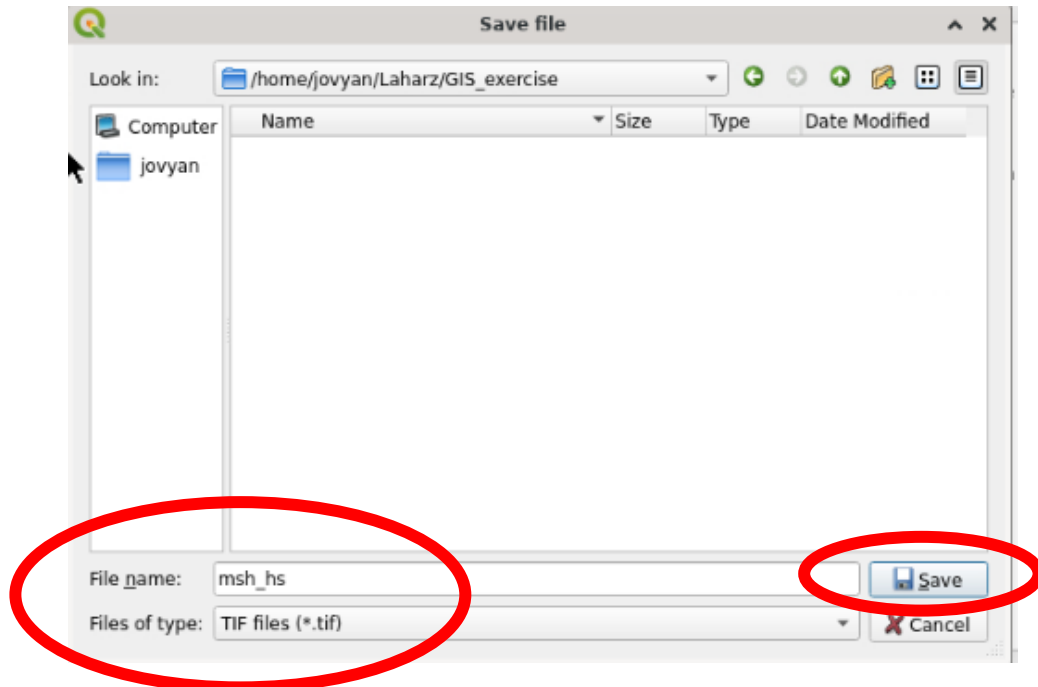
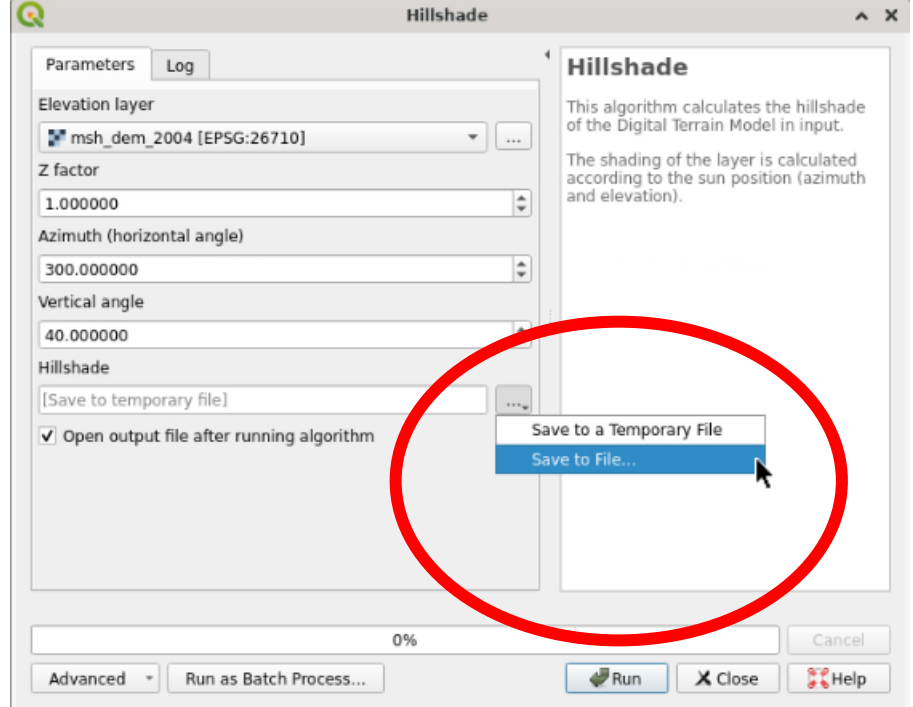
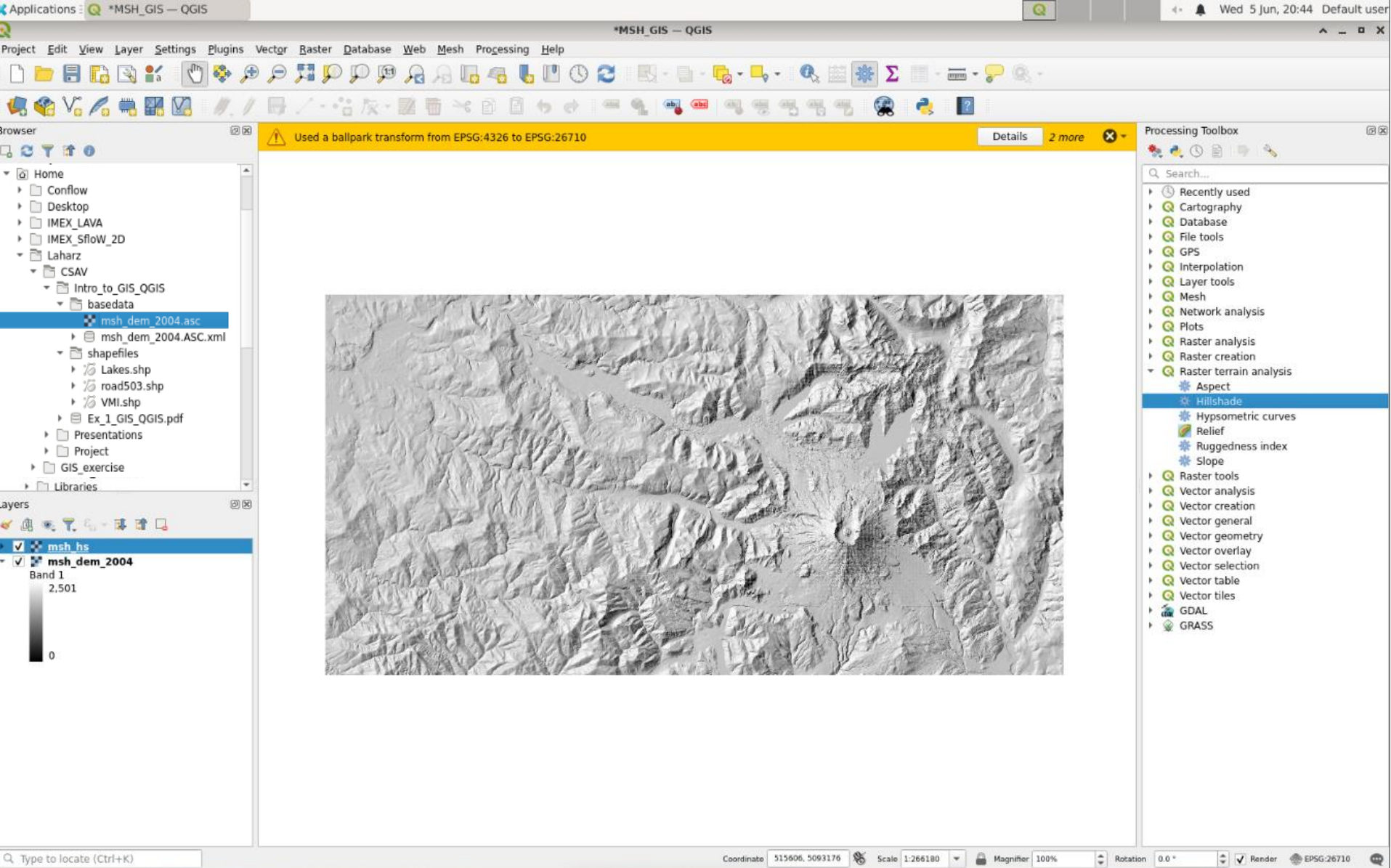


Fig. 28.32 Azimuth and vertical angle

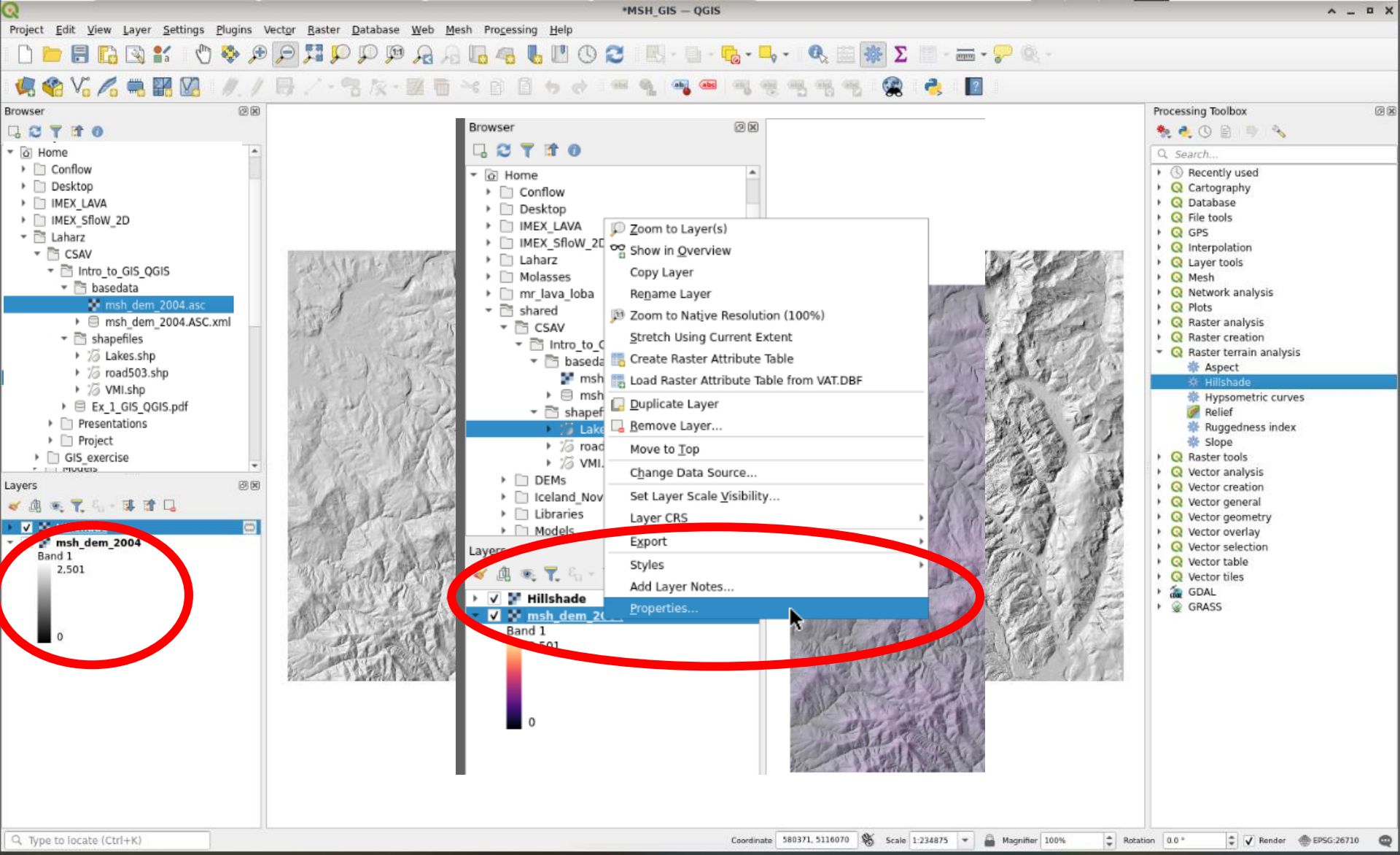
Let's make a hillshade

- Click the 3 dot menu next to the Hillshade filename box and select “Save to File”
- Navigate to Laharz/GIS_exercise
- Enter a filename. I used “msh_hs”
- Click “Save”
- Click “Run”

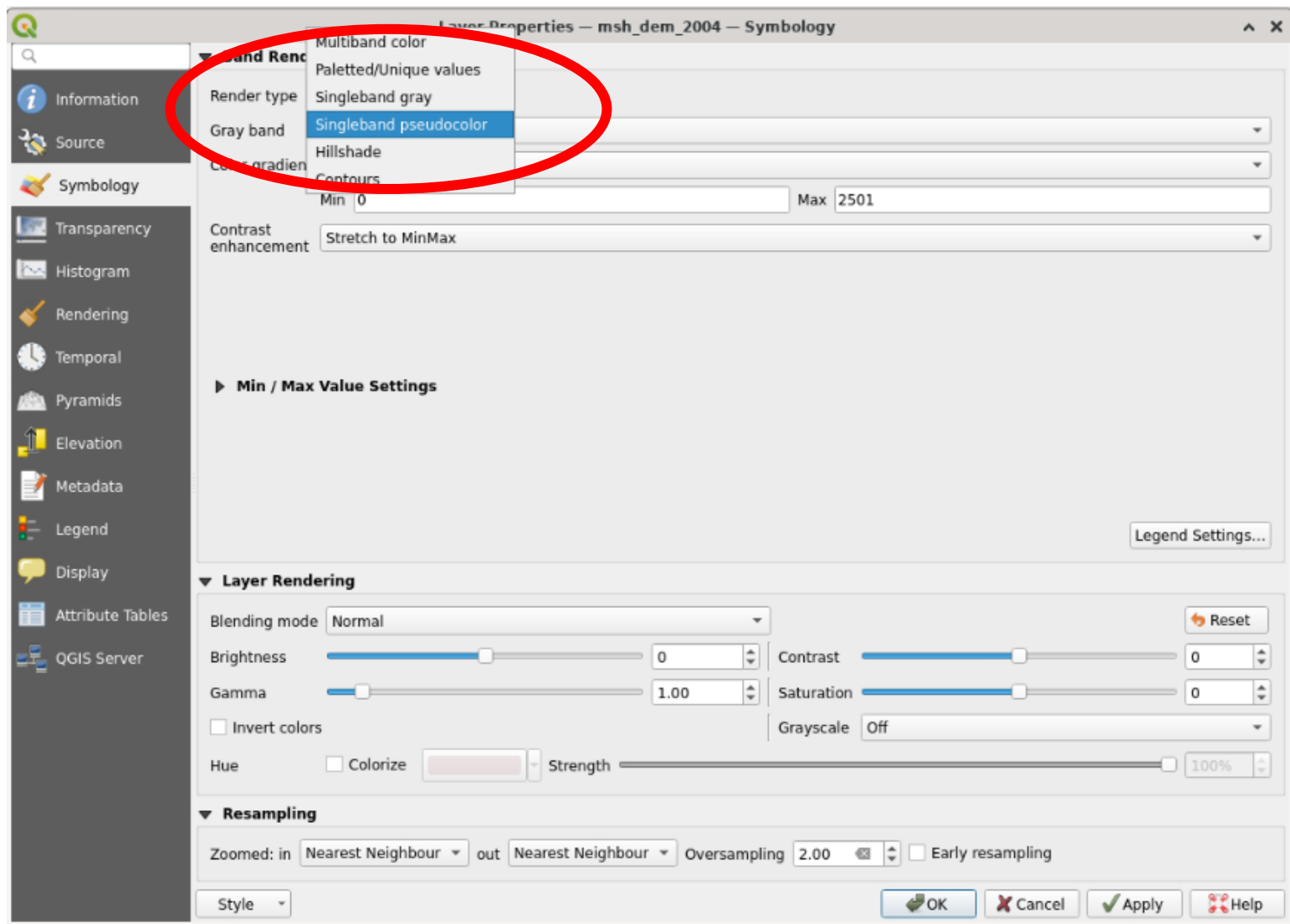




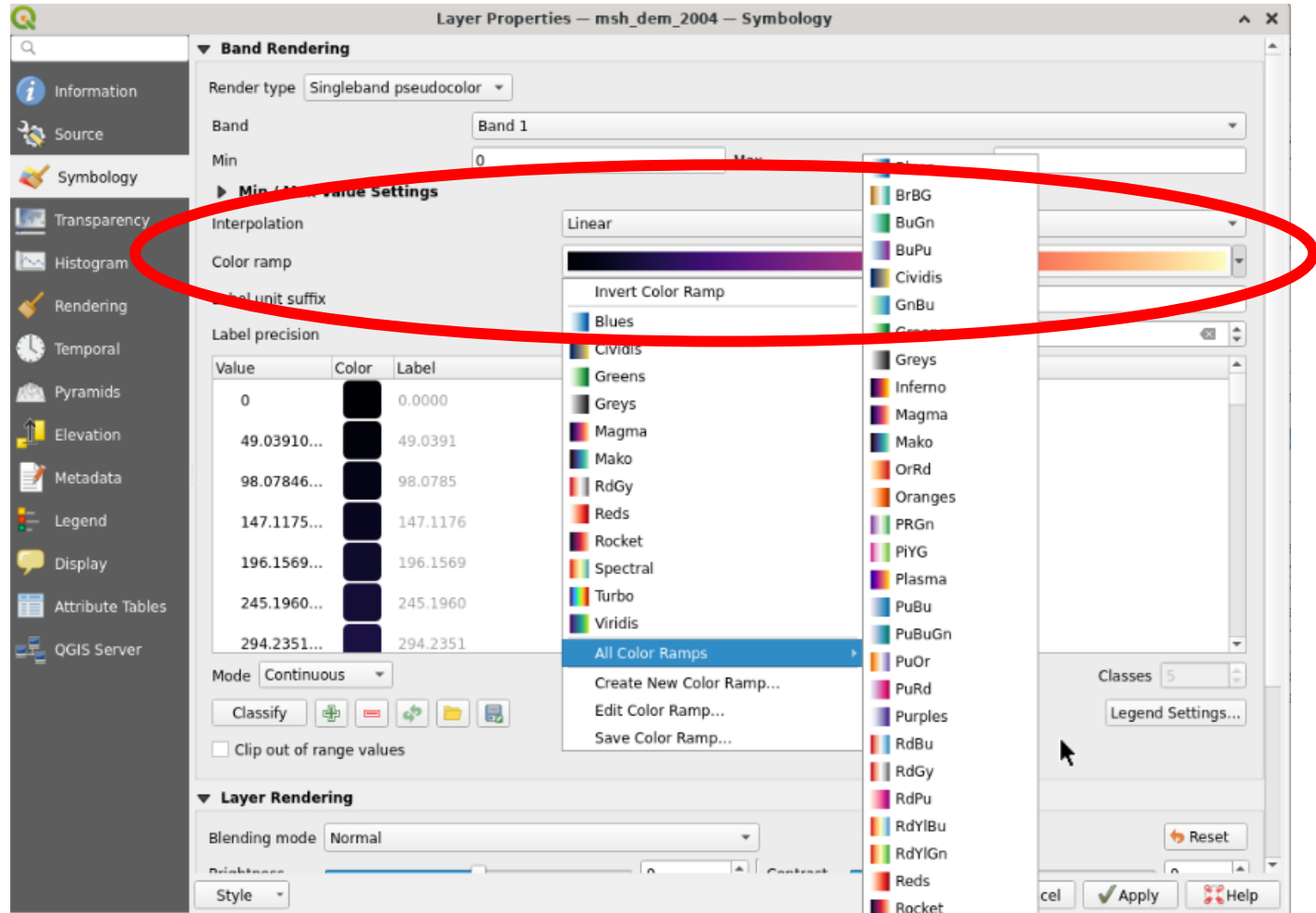
- We can change the symbology of the DEM and the transparency of the hillshade if we want
- Hillshades (shaded relief) use shades of grey 0-254. DEMs use a stretched color spectrum from lowest to highest elevation



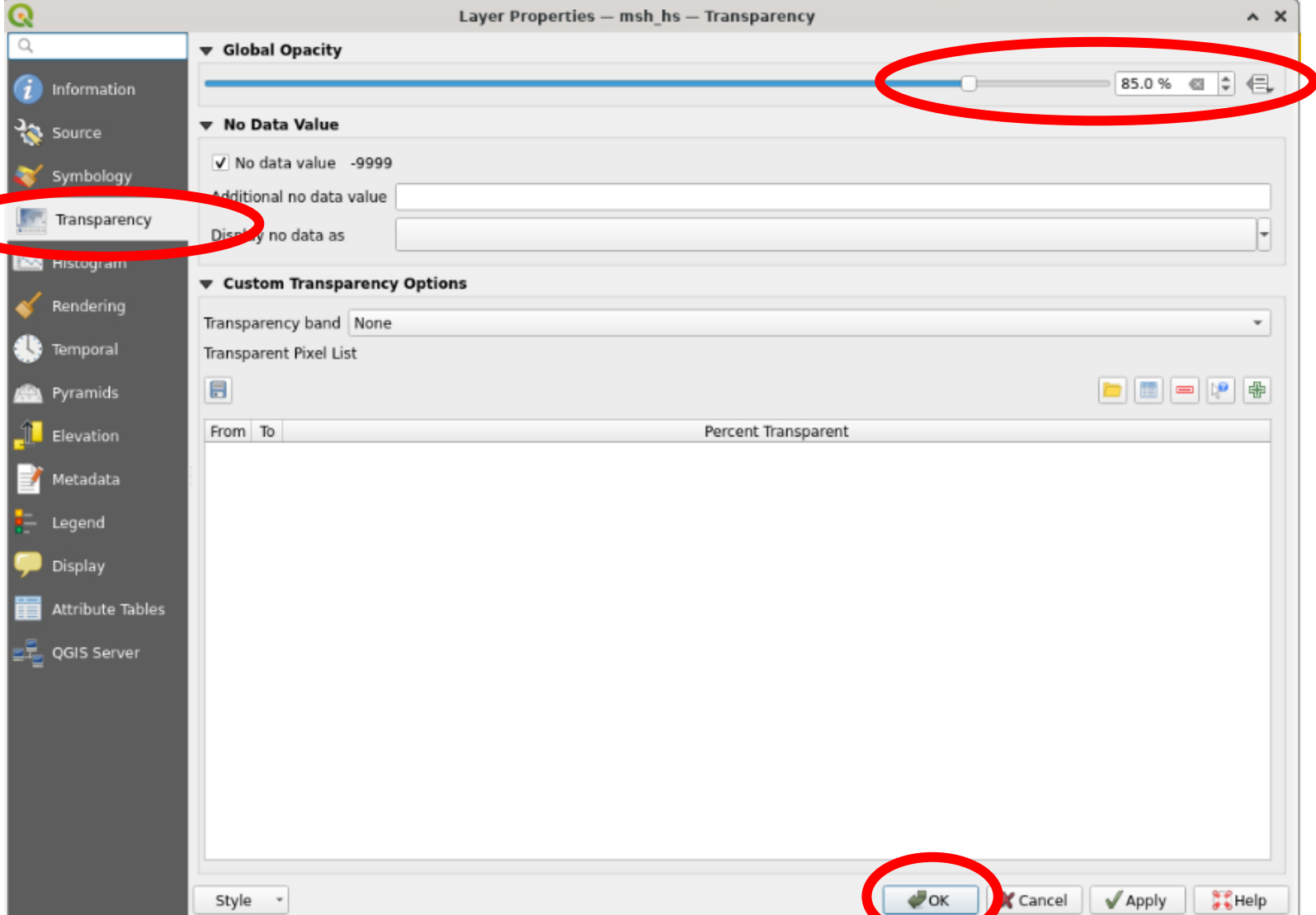
- Double click on the msh_dem_2004 layer to open the layer properties
- OR right click on the “msh_dem_2004” layer and choose “Properties”



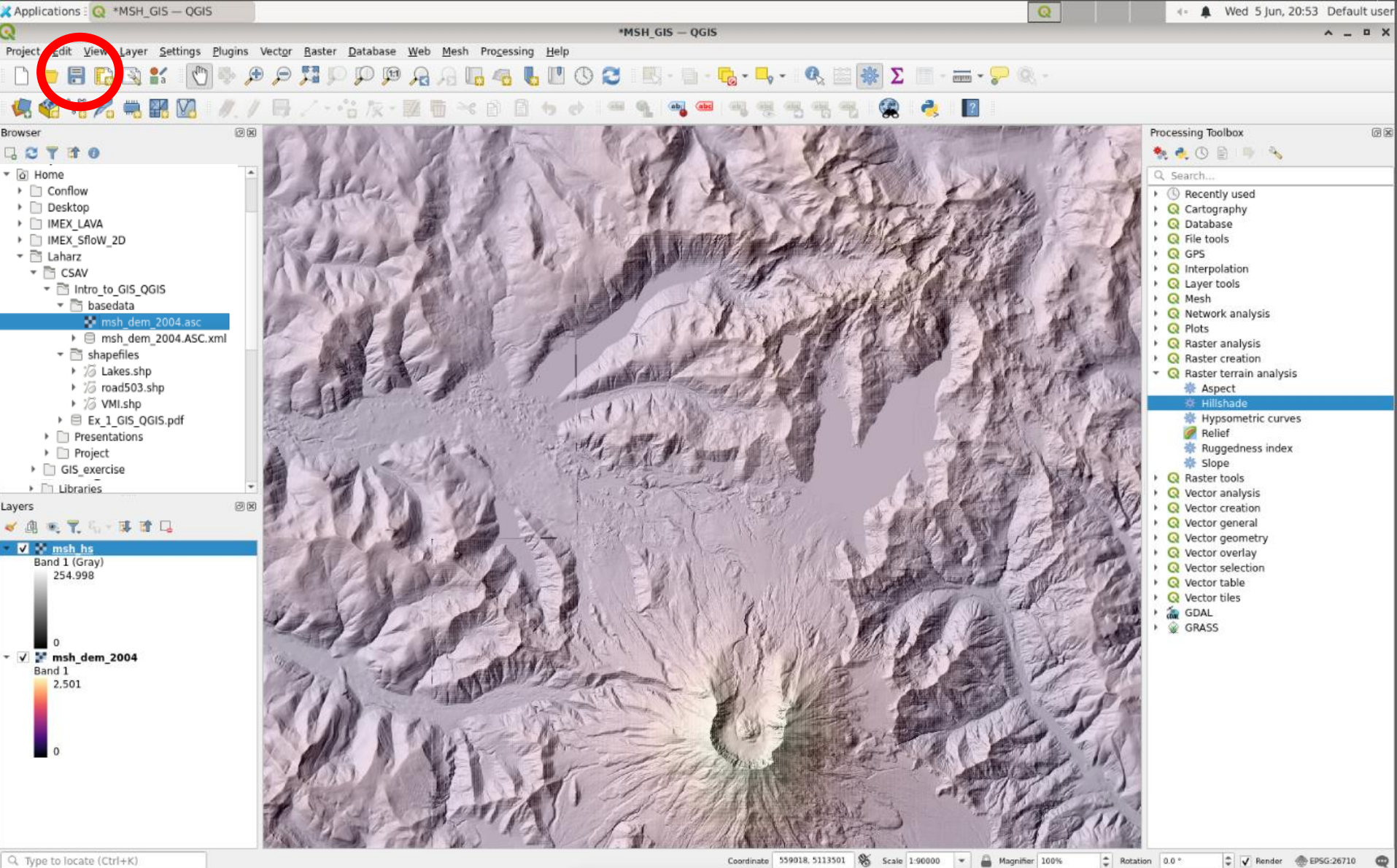
- Change the “Render type” to “Singleband pseudocolor”



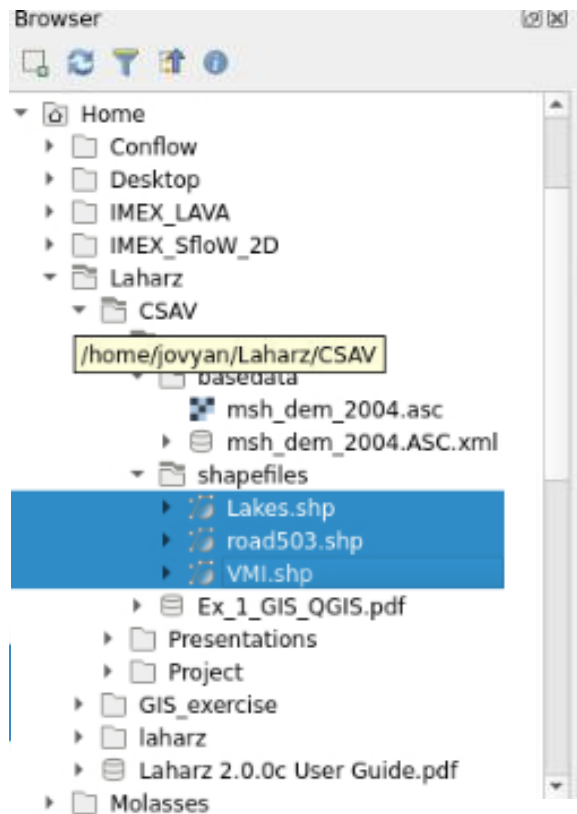
- Choose a color ramp
- I chose the “Magma” color ramp
- Click “Ok”



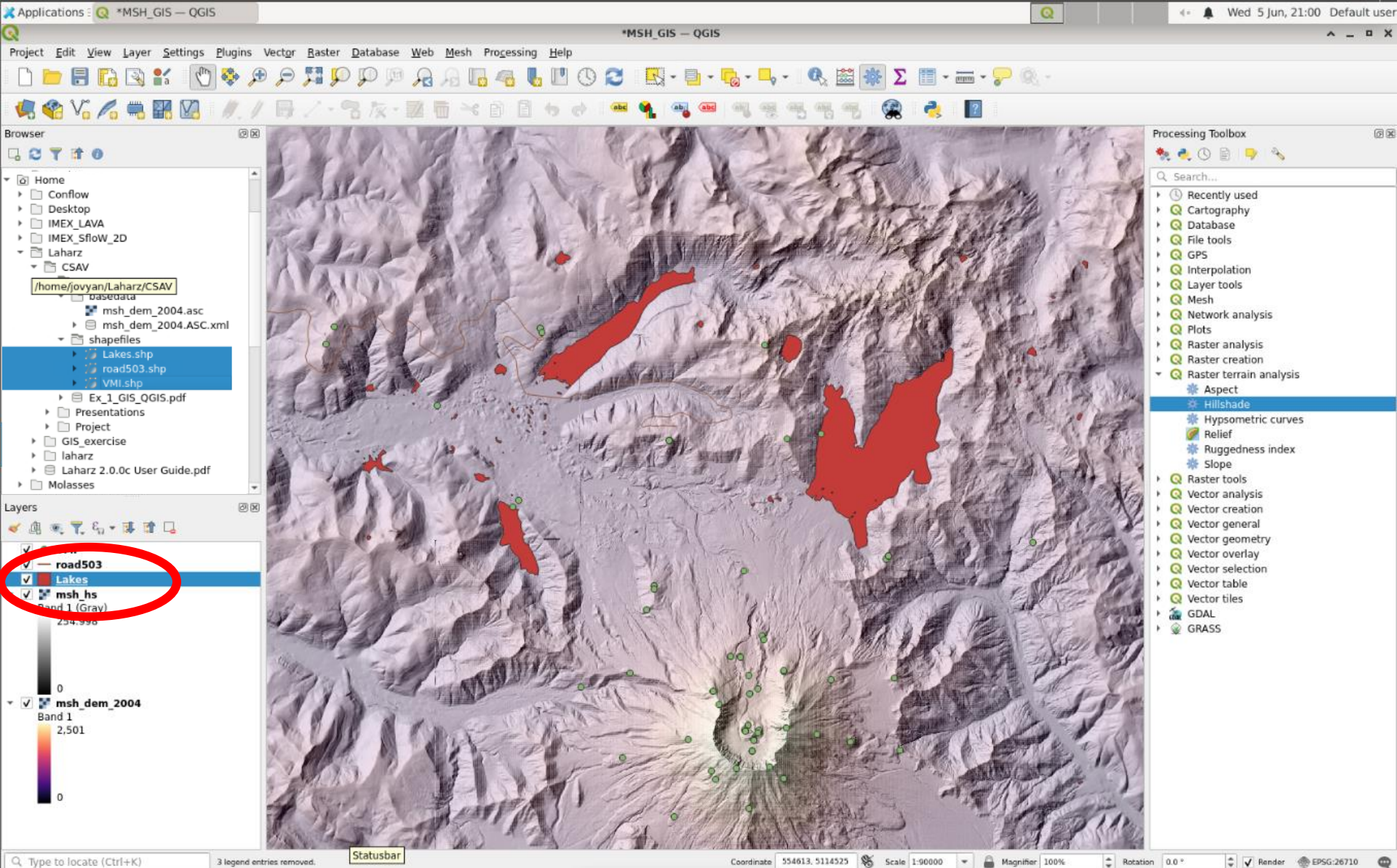
- Now let's make the hillshade slightly transparent
- Right click on "msh_hs" and select "Properties"
- Then select "Transparency" from the left menu
- Change the "Global opacity" to something less than 100% (I chose 85%)
- Click "OK"



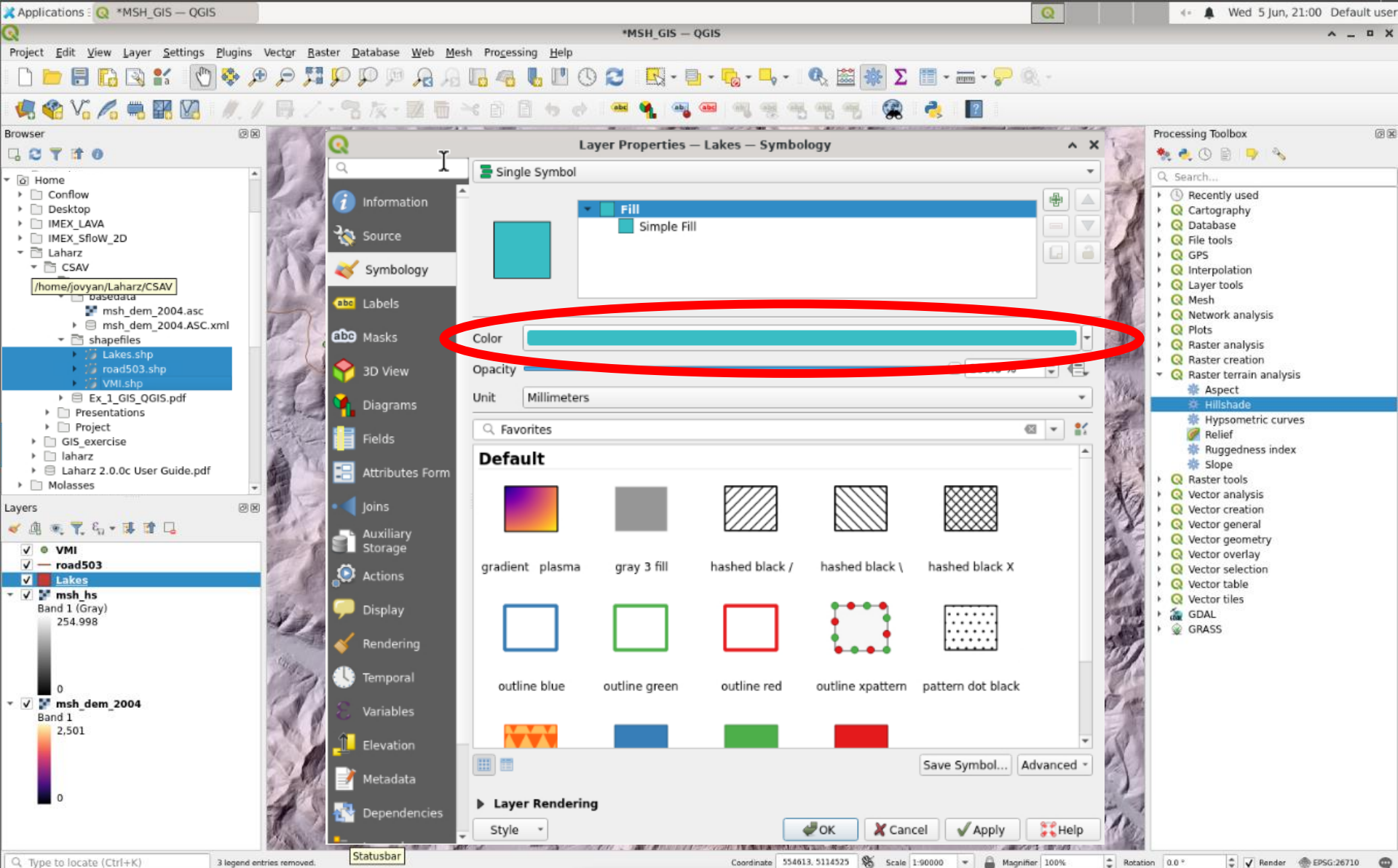
- Now's a good time to save your project!
- Click the “Save” icon in the upper left



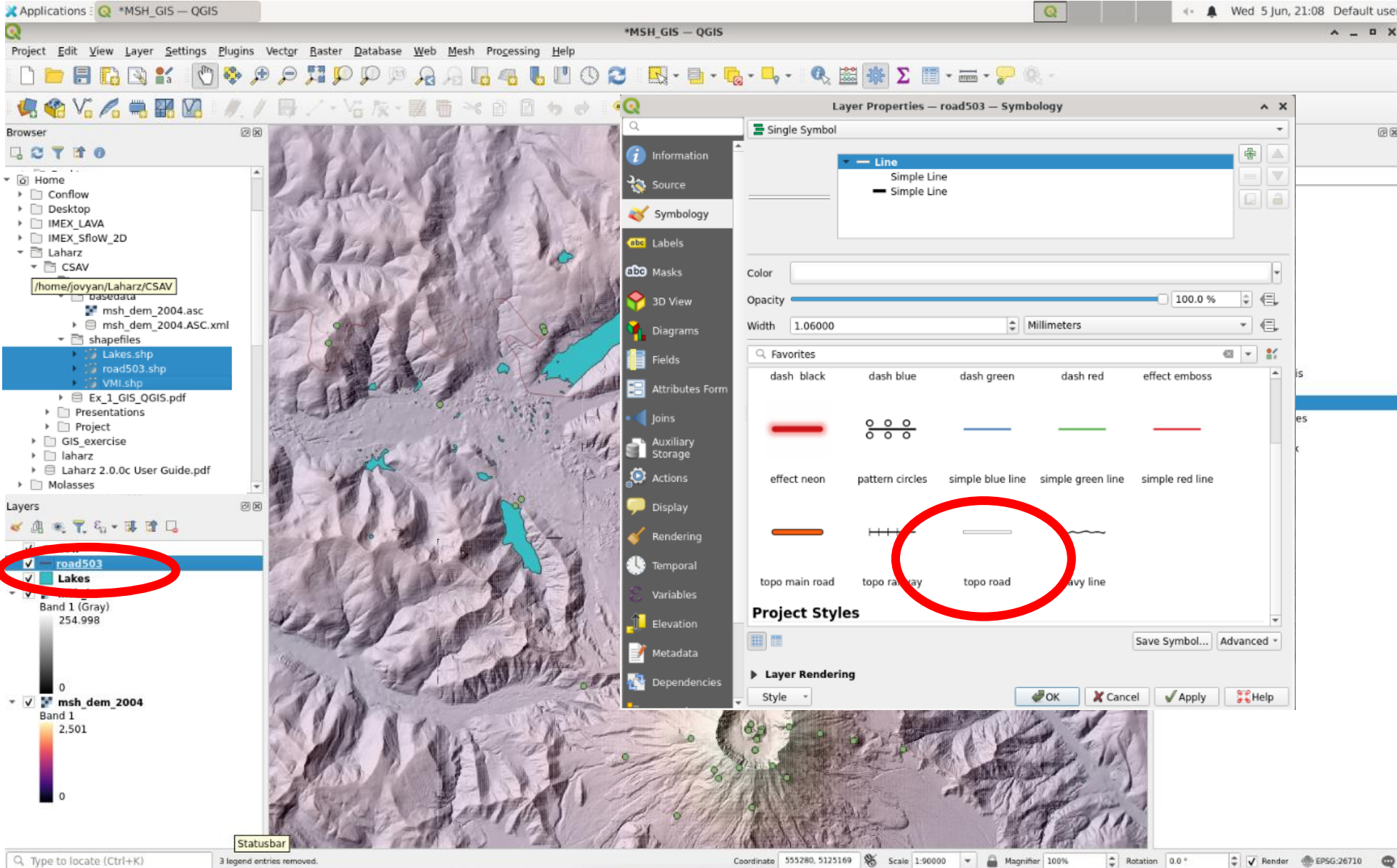
- Add data: Lakes, road503, and VMI shapefiles
- QGIS warns that these datasets have different coordinate systems
- You can leave the choice of transformations as the defaults



- The default symbology for these layers isn't very nice...
- Let's fix it!
- Double click on the “Lakes” layer to open the Layer Properties

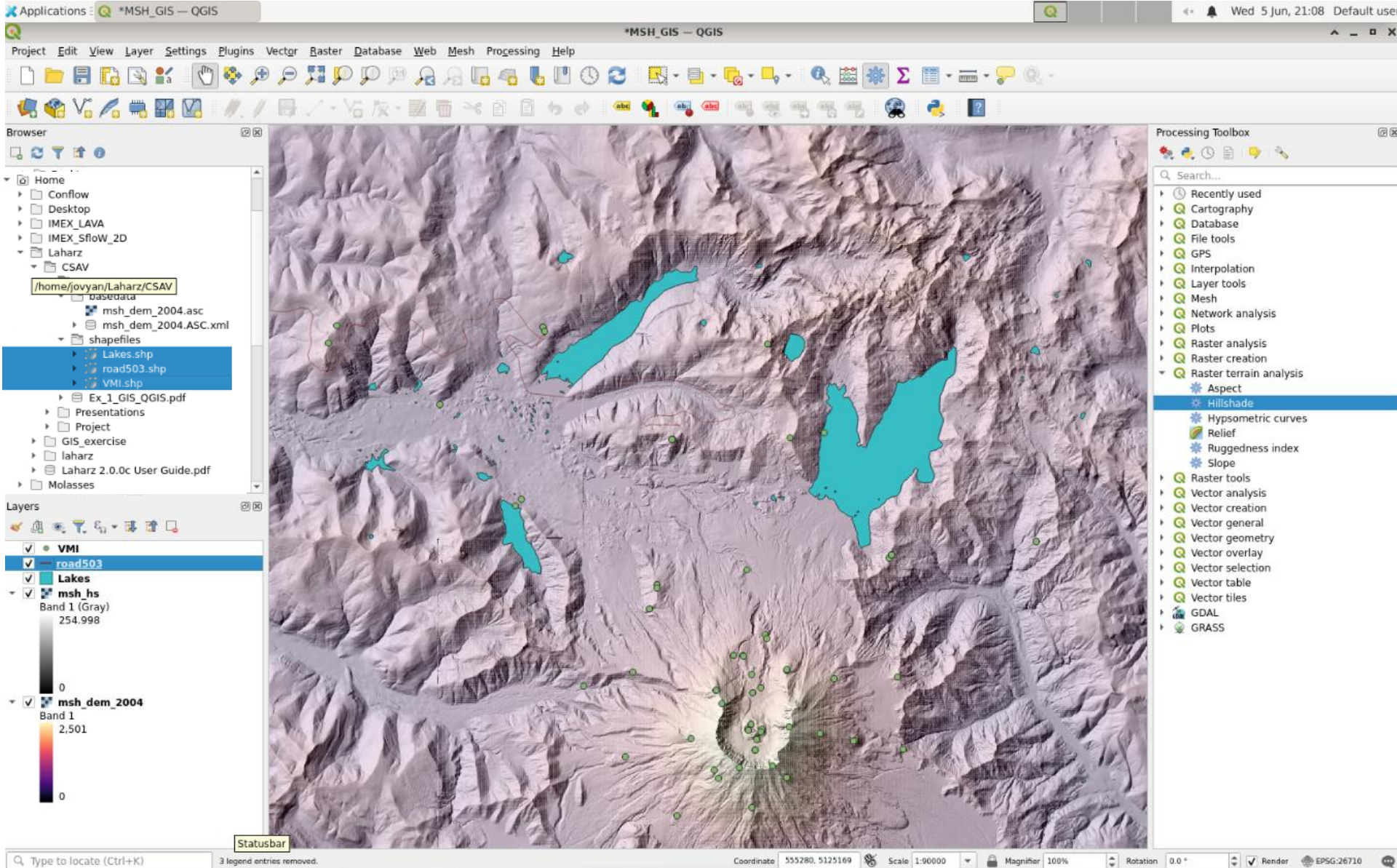


- Change the lake color to a lake color
- Use the drop-down menu labelled “color” and choose a color
- Click “Ok”

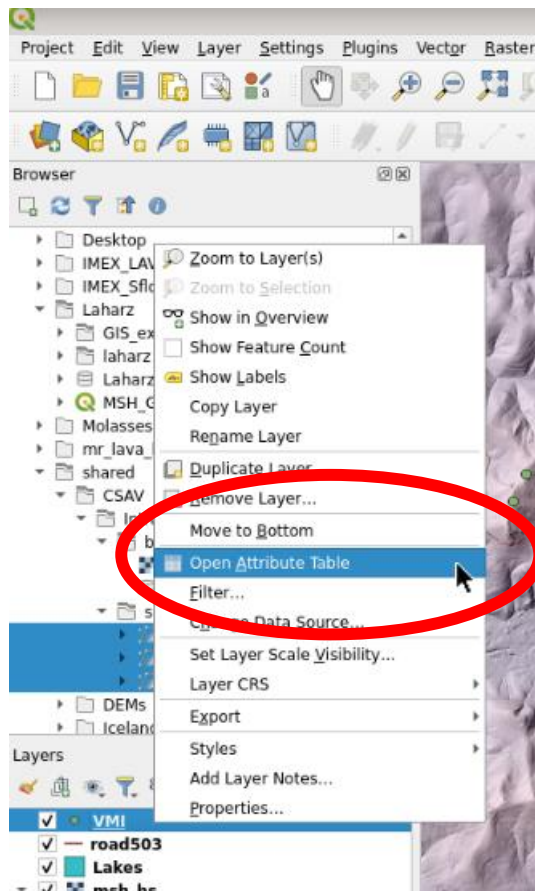


Exercise Goals

- Add Instrument data set and background information for context
- **View Instrument attribute table**
- Select instruments by drawing rectangle
- Select instruments by their attributes
- Assign colors to instrument type
- Add a legend to describe the meaning of colors



- The last layer is “VMI” or “volcano monitoring instruments”. It shows the network of instruments of different types.
- The default symbols are all the same, so we can’t tell what type of instruments these are. Let’s fix that!



VMI — Features Total: 78, Filtered: 78, Selected: 1

	LOSEST_VO	VMI INSTR TYPE	SENSOR	LATITUDE	LONGITUDE	ELEV	STATE	LAND_MGR	LAND UNIT	WILDERNESS	TLMTRY	PROCESSING	START_DATE	CLOSE_DATE	ICHI3NETWC	Short_Desc
1	unt St. He...	31.000000	Mark L10 Ge...	46.211667	-122.251667	1492.000000	WA	FS	Mount St. He...	NULL	1.000000	CVO	7/18/05	NULL	C...	Acoustic Flow Monitor
2	unt St. He...	31.000000	Mark L10 Ge...	46.241389	-122.190556	1146.000000	WA	FS	Mount St. He...	NULL	1.000000	CVO	6/12/95	NULL	C...	Acoustic Flow Monitor
3	unt St. He...	31.000000	Mark L10 Ge...	46.223443	-122.183342	1585.000000	WA	FS	Mount St. He...	NULL	1.000000	CVO	6/12/95	NULL	C...	Acoustic Flow Monitor
4	unt St. He...	31.000000	Mark L10 Ge...	46.284722	-122.303889	661.000000	WA	FS	Mount St. He...	NULL	1.000000	CVO	6/12/95	NULL	C...	Acoustic Flow Monitor
5	unt St. He...	36.000000	anemometer	46.247784	-122.083017	1024.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	10/1/98	NULL	C...	Anemometer
6	unt St. He...	7.000000	Malin Boreho...	46.303300	-122.264800	990.000000	WA	FS	Mount St. He...	NULL	2.000000	PNSN	9/13/07	NULL	U...	Borehole Seismome...
7	unt St. He...	7.000000	Malin Boreho...	46.244700	-122.136700	1219.000000	WA	FS	Mount St. He...	NULL	2.000000	PNSN	7/26/07	NULL	U...	Borehole Seismome...
8	unt St. He...	17.000000	Gladwin Bore...	46.303300	-122.264800	990.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO/UNAVCO	9/12/07	NULL	U...	Borehole Strainmeter
9	unt St. He...	17.000000	Gladwin Bore...	46.244700	-122.136700	1219.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO/UNAVCO	7/25/07	NULL	U...	Borehole Strainmeter
10	unt St. He...	11.000000	Pinnacle 5000	46.199733	-122.185567	2079.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	11/17/05	5/29/2006	C...	Borehole Tiltmeter
11	unt St. He...	11.000000	Applied Geo...	46.201400	-122.189800	2127.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	10/29/07	NULL	C...	Borehole Tiltmeter
12	unt St. He...	11.000000	Applied Geo...	46.219200	-122.195900	1630.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	9/23/06	NULL	C...	Borehole Tiltmeter
13	unt St. He...	11.000000	Lily Borehole...	46.231479	-122.226948	1183.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	9/22/06	NULL	U...	Borehole Tiltmeter
14	unt St. He...	11.000000	Lily Borehole...	46.180100	-122.189800	2091.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	2/1/05	NULL	U...	Borehole Tiltmeter
15	unt St. He...	11.000000	Lily Borehole...	46.210322	-122.202348	2121.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	10/14/04	NULL	U...	Borehole Tiltmeter
16	unt St. He...	11.000000	Pinnacle 5000	46.210820	-122.185950	1712.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	12/8/07	NULL	C...	Borehole Tiltmeter
17	unt St. He...	11.000000	Pinnacle 5000	46.219200	-122.195900	1630.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	9/5/06	NULL	C...	Borehole Tiltmeter

Show All Features

- Right click on “VMI” and choose “Open Attribute Table”
- We can see all kinds of information about the sensors, including the sensor type (“Short_Desc” column all the way to the right)

Applications: QGIS3 *MSH_GIS - QGIS

Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help

Browser

- Home
 - Conflow
 - Desktop
 - IMEX_LAVA
 - IMEX_Sflow_2D
 - Laharz
 - CSAV
 - /home/jovyan/Laharz/CSAV
 - base data
 - msh_dem_2004.asc
 - msh_dem_2004.ASC.xml
 - shapefiles
 - Lakes.shp
 - road503.shp
 - VMI.shp
 - Ex_1_GIS_QGIS.pdf
 - Presentations
 - Project
 - GIS_exercise
 - laharz
 - Laharz 2.0.0c User Guide.pdf
 - Molasses

Layers

- ☒ VMI
- ☒ road503
- ☒ Lakes
- ☒ msh_hs
 - Band 1 (Gray)
 - 254.998
- ☒ msh_dem_2004
 - Band 1
 - 2.501

13 features selected on layer VMI.

VMI — Features Total: 78, Filtered: 78, Selected: 13

	IPONSOF	OPERATOR	VO	STATION	SITE	CLOSEST_VO	VNI	INSTR_TYPE	SENSOR	LATITUDE	LONGITUDE
17	USGS	CVO	CVO	WOGU	West of Guac...	Mount St. He...	...	11.000000	Pinnacle 5000	46.219200	-122.195
18	USGS	CVO	CVO	NDM	North Dome	Mount St. He...	...	11.000000	Pinnacle 5000	46.201400	-122.189
19	USGS	CVO	CVO	Sept L...	September L...	Mount St. He...	...	11.000000	Pinnacle 5000	46.200280	-122.190
20	USGS	CVO	CVO	Brutus	Brutus	Mount St. He...	...	51.000000	Canon Digital...	46.200667	-122.179
21	USGS	CVO	CVO	SRIM	South Rim	Mount St. He...	...	51.000000	Olympus C30...	46.190252	-122.194
22	USGS	CVO	CVO	SHOE	Shoestring N...	Mount St. He...	...	51.000000	Olympus C30...	46.190639	-122.181
23	USGS	CVO	CVO	NWRim	Northwest Rim	Mount St. He...	...	51.000000	Canon Digital...	46.206880	-122.200
24	USGS	CVO	CVO	GUAC	Guacamole	Mount St. He...	...	51.000000	Canon Digital...	46.219060	-122.192
25	USGS	CVO	CVO	Sugar...	Sugarbowl	Mount St. He...	...	51.000000	Olympus C30...	46.215556	-122.176
26	USGS	CVO	CVO	Spine ...	Spine 5	Mount St. He...	...	52.000000	Canon Digital...	46.194861	-122.188
27	USGS	CVO	CVO	Spine ...	Spine 5	Mount St. He...	...	52.000000	Canon Digital...	46.194861	-122.188
28	USGS	CVO	CVO	Goodi...	Spine 5	Mount St. He...	...	52.000000	Canon Digital...	46.194861	-122.188

Show All Features

Processing Toolbox

- Recently used
- Cartography
- Database
- File tools
- GPS
- Interpolation
- Layer tools
- Mesh
- Network analysis
- Plots
- Raster analysis
- Raster creation
- Raster terrain analysis
- Aspect
 - Hillshade
 - Hypsometric curves
 - Relief
 - Ruggedness index
 - Slope
- Raster tools
 - Vector analysis
 - Vector creation
 - Vector general
 - Vector geometry
 - Vector overlay
 - Vector selection
 - Vector table
 - Vector tiles
- GDAL
- GRASS

Type to locate (Ctrl+K)

16264 Scale 1:90000 Magnifier 100% Rotation 0.0° Render EPSG:26710

- When you select rows in the attribute table (Ctrl+Click the row number), you will see those instruments turn yellow on the map
- Close the attribute table

Exercise Goals

- Add Instrument data set and background information for context
- View Instrument attribute table
- **Select instruments by a drawing rectangle**
- Select instruments by their attributes
- Assign colors to instrument type
- Add a legend to describe the meaning of colors

Applications: QGIS3

Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help

Processing Toolbox

Search...

- Recently used
- Cartography
- Database
- File tools
- GPS
- Interpolation
- Layer tools
- Mesh
- Network analysis
- Plots
- Raster analysis
- Raster creation
- Raster terrain analysis
 - Aspect
 - Hillshade
 - Hypsometric curves
 - Relief
 - Ruggedness index
 - Slope
- Raster tools
- Vector analysis
- Vector creation
- Vector general
- Vector geometry
- Vector overlay
- Vector selection
- Vector table
- Vector tiles
- GDAL
- GRASS

VMI — Features Total: 78, Filtered: 78, Selected: 19

ID	SPONSOR	OPERATOR	VO	STATION	SITE	CLOSEST_VO	VNI	INSTR_TYPE	S
11	USGS	CVO	CVO	NDML	North Dome ...	Mount St. He...	...	11.000000	Appli
12	USGS	CVO	CVO	WOGU	West of Guac...	Mount St. He...	...	11.000000	Appli
13	NSF	UNAVCO	CVO	P691	Studebaker R...	Mount St. He...	...	11.000000	Lily E
14	NSF	UNAVCO	CVO	P690	South Ridge A	Mount St. He...	...	11.000000	Lily E
15	NSF	UNAVCO	CVO	P693	Northwest D...	Mount St. He...	...	11.000000	Lily E
16	USGS	CVO	CVO	GLAC	Glacier	Mount St. He...	...	11.000000	Pinn
17	USGS	CVO	CVO	WOGU	West of Guac...	Mount St. He...	...	11.000000	Pinn
18	USGS	CVO	CVO	NDM	North Dome	Mount St. He...	...	11.000000	Pinn
19	USGS	CVO	CVO	Sept L...	September L...	Mount St. He...	...	11.000000	Pinn
20	USGS	CVO	CVO	Brutus	Brutus	Mount St. He...	...	51.000000	Canc
21	USGS	CVO	CVO	SRIM	South Rim	Mount St. He...	...	51.000000	Olym
22	USGS	CVO	CVO	SHOE	Shoestring N...	Mount St. He...	...	51.000000	Olym
23	USGS	CVO	CVO	NWRim	Northwest Rim	Mount St. He...	...	51.000000	Canc
24	USGS	CVO	CVO	GUAC	Guacamole	Mount St. He...	...	51.000000	Canc

Show All Features

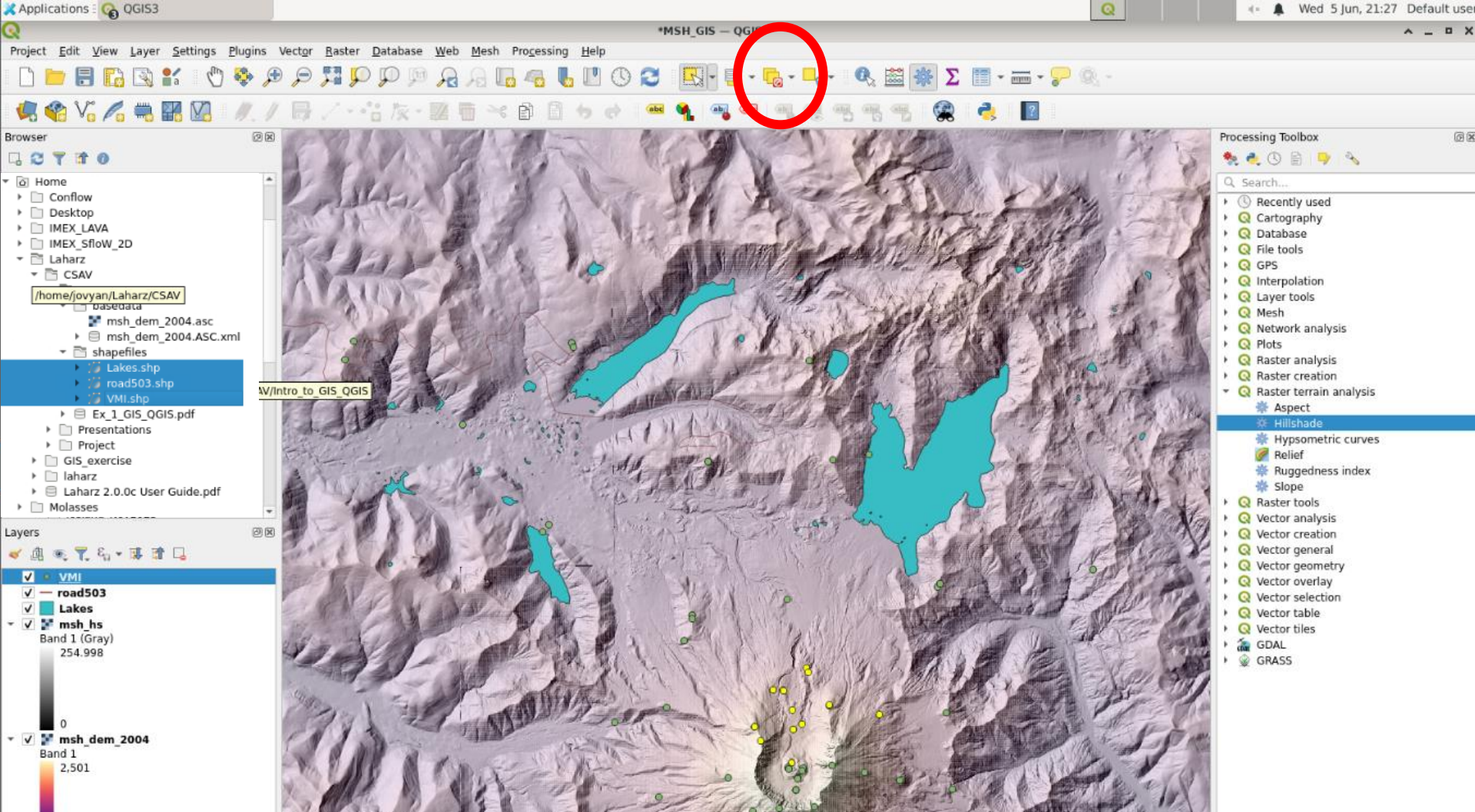
Layers

- VMI
- road503
- Lakes
- msh_hs
 - Band 1 (Gray) 254.998
- msh_dem_2004
 - Band 1 2.501

0 features selected on layer VMI

Coordinate 566091, 5115430 Scale 1:90000 Magnifier 100% Rotation 0.0° Render EPSG:26710

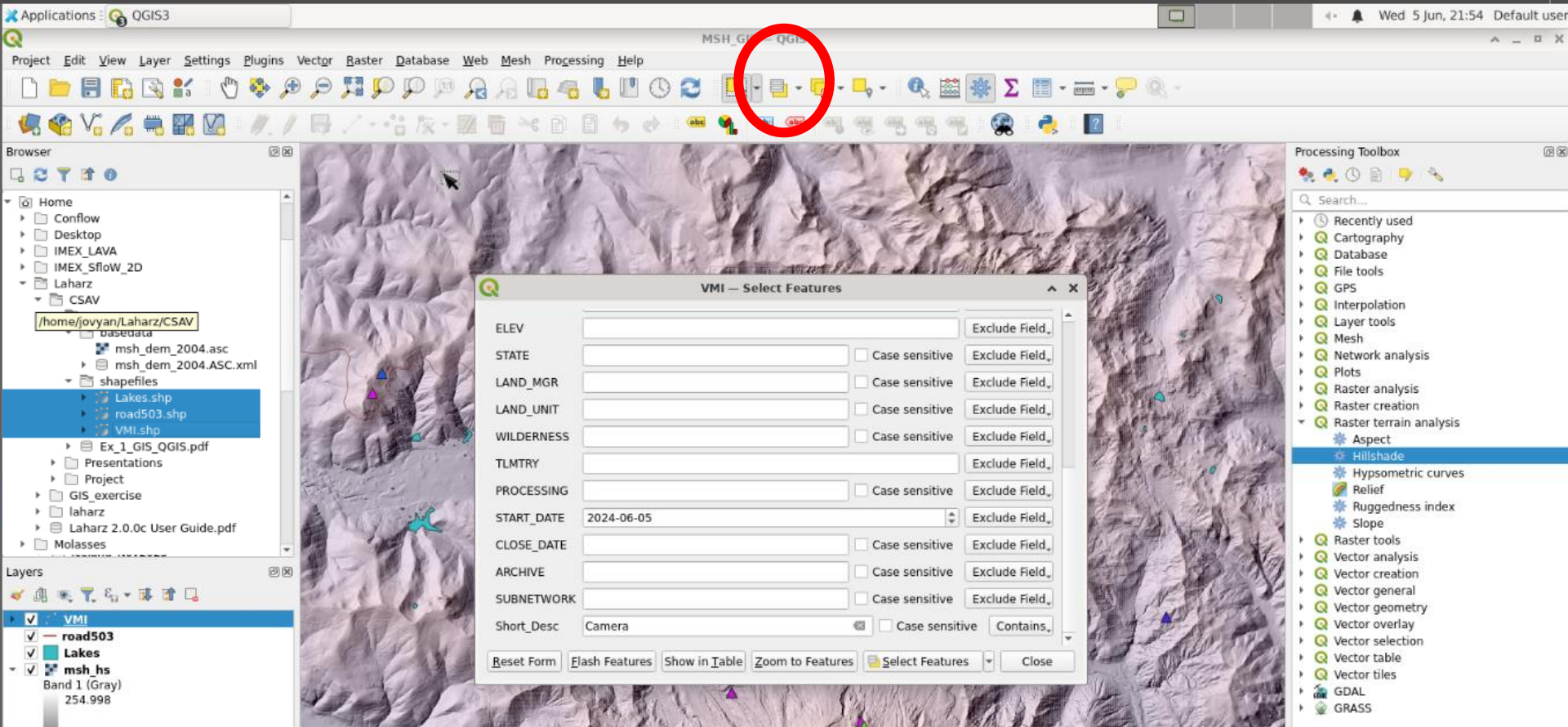
- Choose the “Select” tool from the menu
- Draw a rectangle around some of the VMI stations
- The selected stations will highlight in yellow
- If you open the attribute table you will see the selected stations highlighted



- To clear the selection, click the “Deselect features from all layers” button

Exercise Goals

- Add Instrument data set and background information for context
- View Instrument attribute table
- Select instruments by a drawing rectangle
- **Select instruments by their attributes**
- Assign colors to instrument type
- Add a legend to describe the meaning of colors

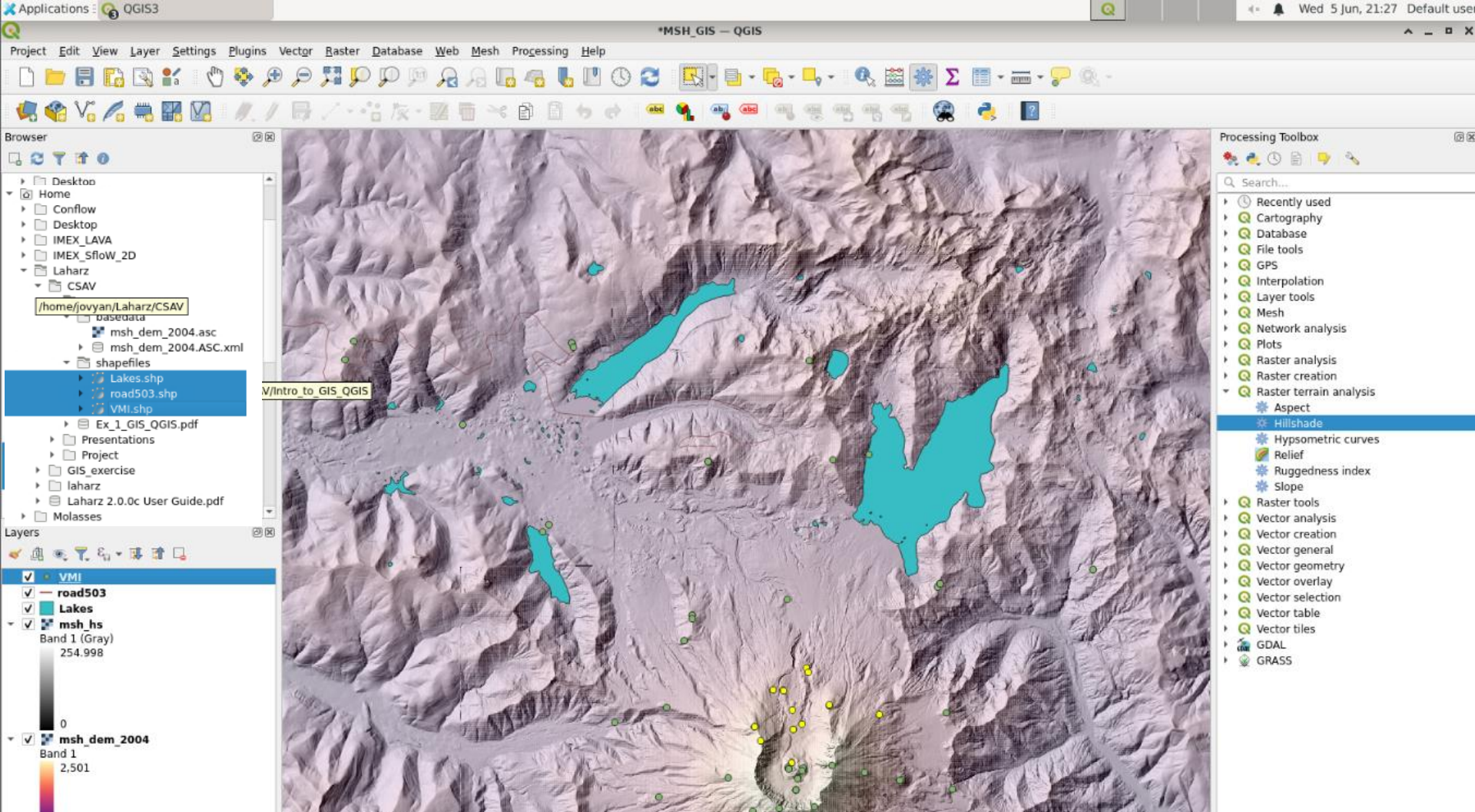


- We can query the attribute table to select certain features based on data from the table
- Select the “Select features by value” tool
- Scroll down until you see “Short_Desc”
- Change the button that says “Exclude Field” to “Contains”
- Type “Camera” in the text box
- Click “Select Features”

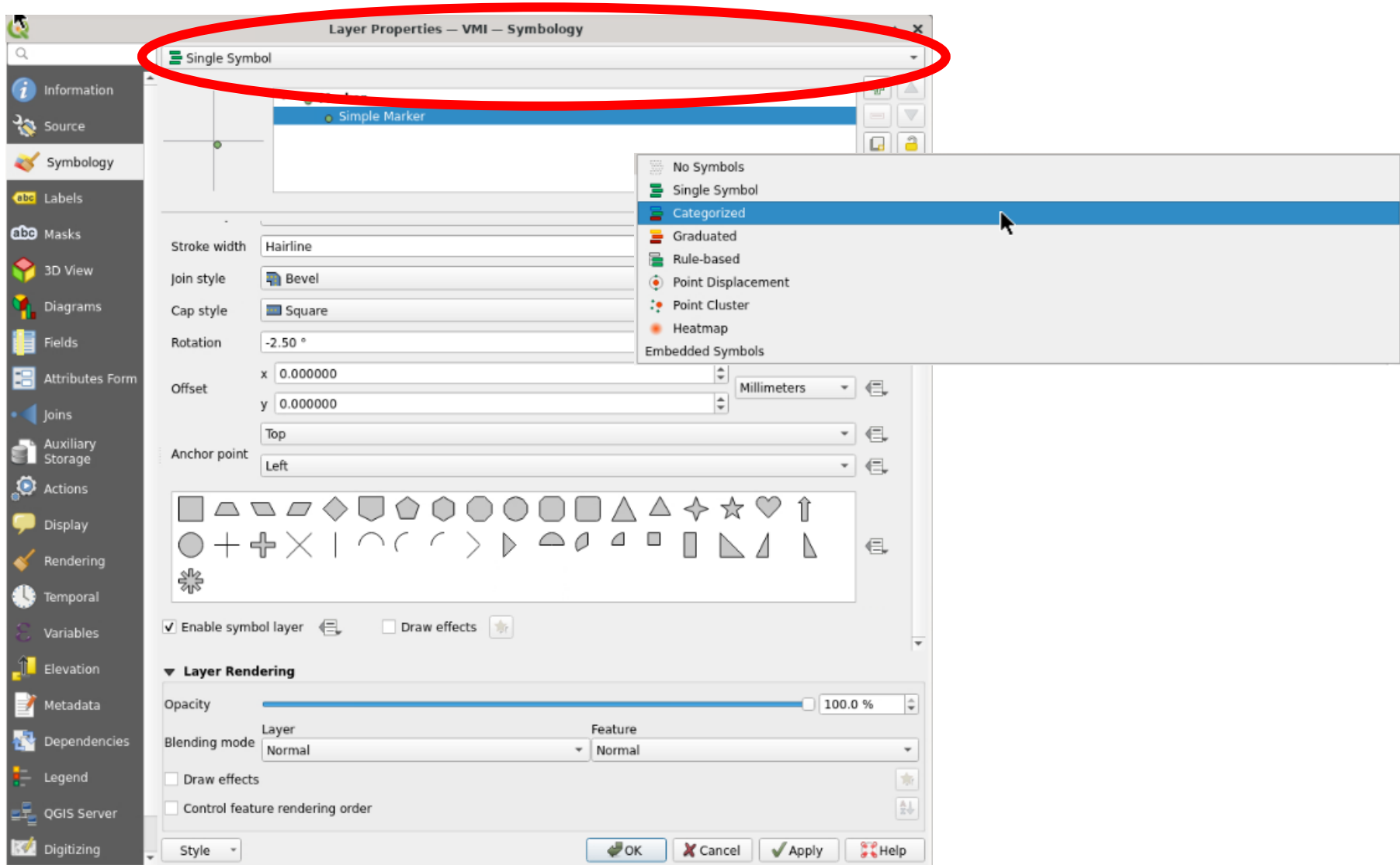


Exercise Goals

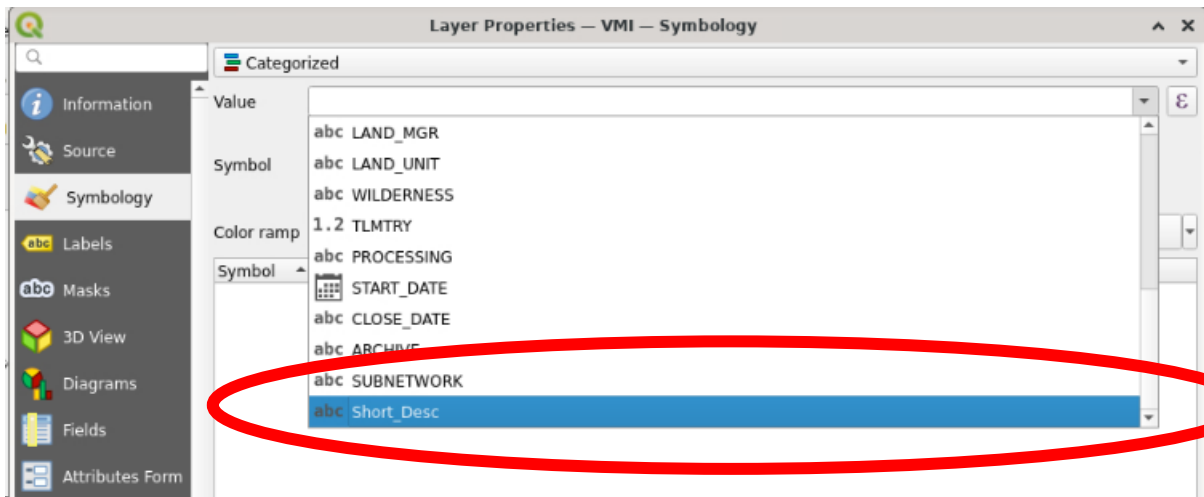
- Add Instrument data set and background information for context
- View Instrument attribute table
- Select instruments by a drawing rectangle
- Select instruments by their attributes
- **Assign colors to instrument type**
- Add a legend to describe the meaning of colors



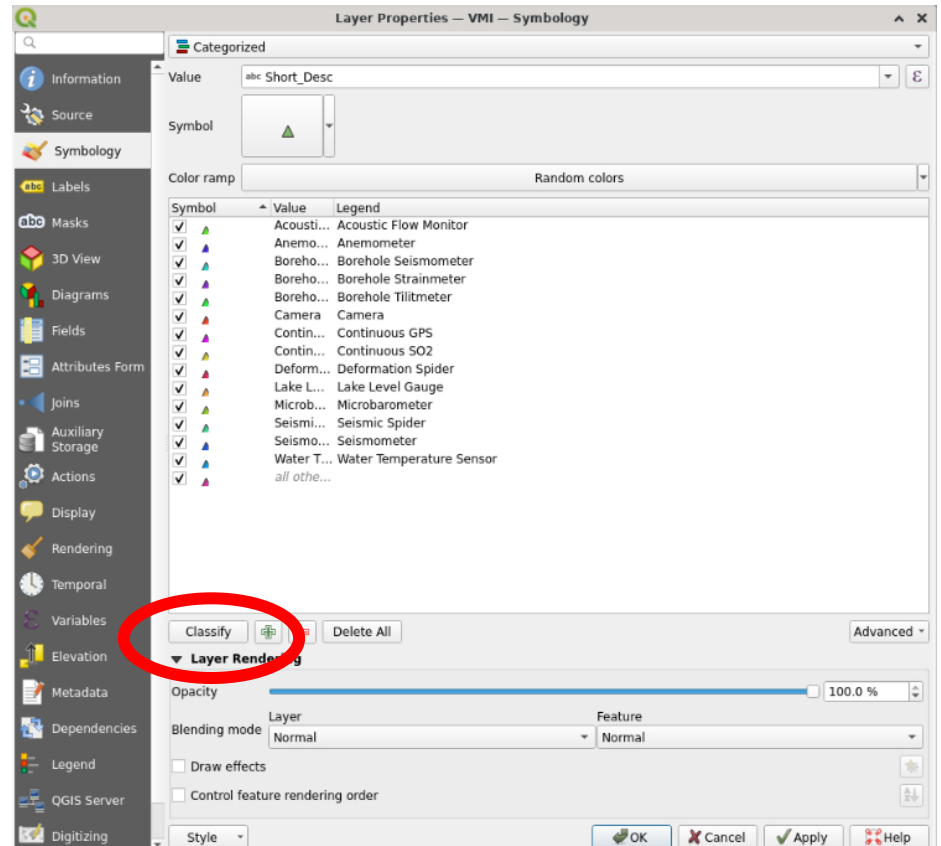
- We can use the attribute table to assign different symbols to the VMI layer so we can more easily tell what kind of stations these are
- Double click the VMI layer and open the Layer Properties

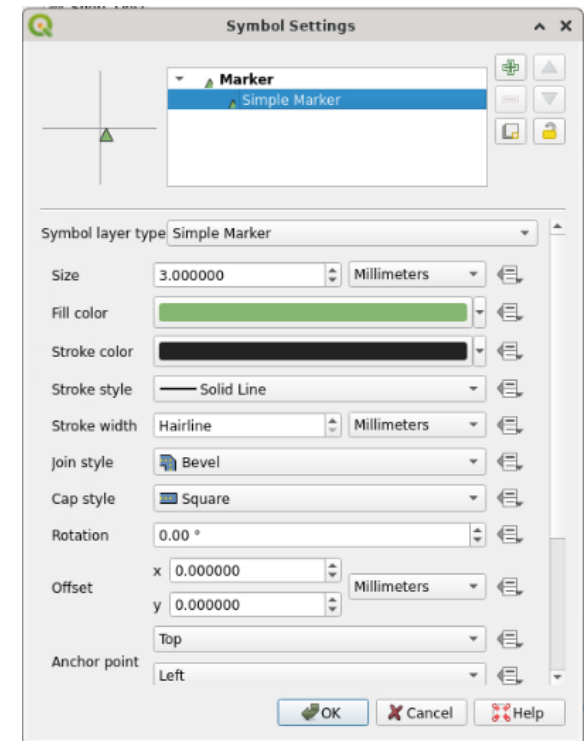
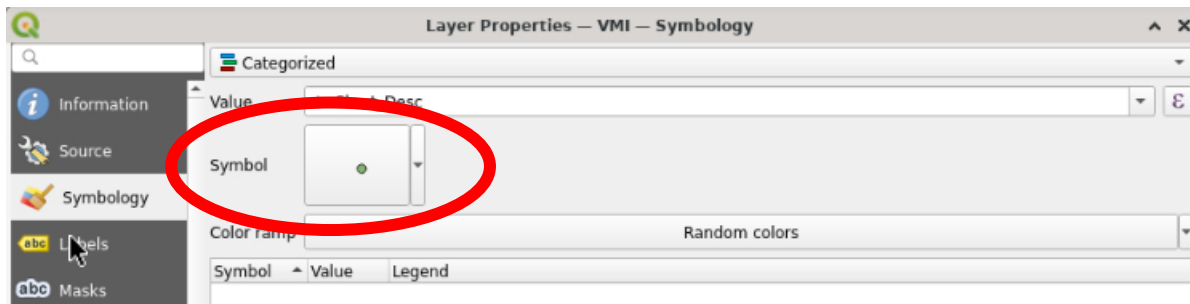


- Change from “Single symbol” to “Categorized”

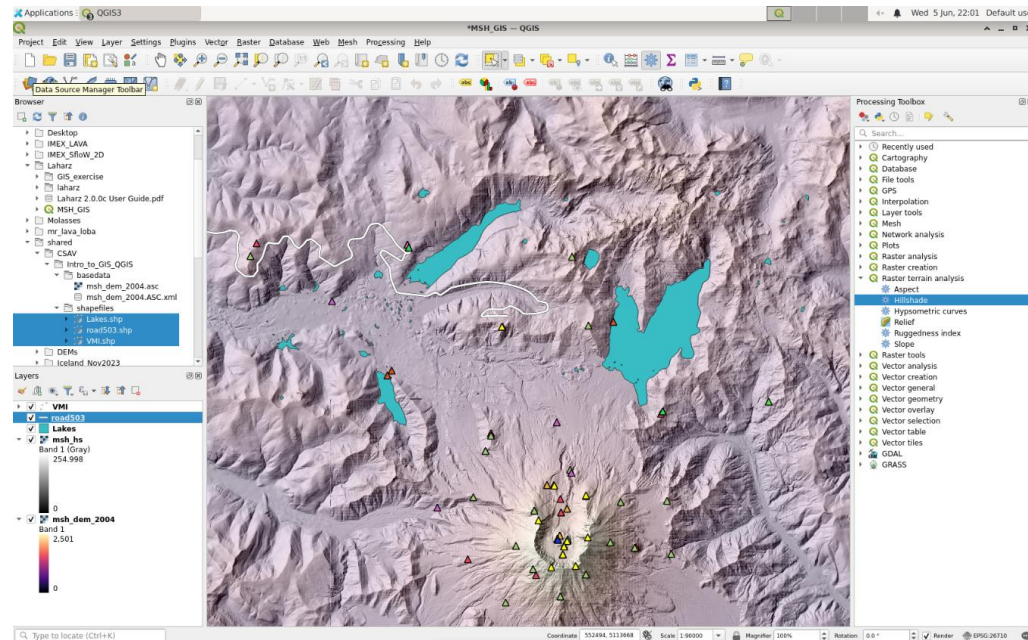


- Change the “Value” to the column you want to use for the display. We want to know what kind of stations these are, so choose “Short_desc”.
- Then click “classify” and all the different station types should appear



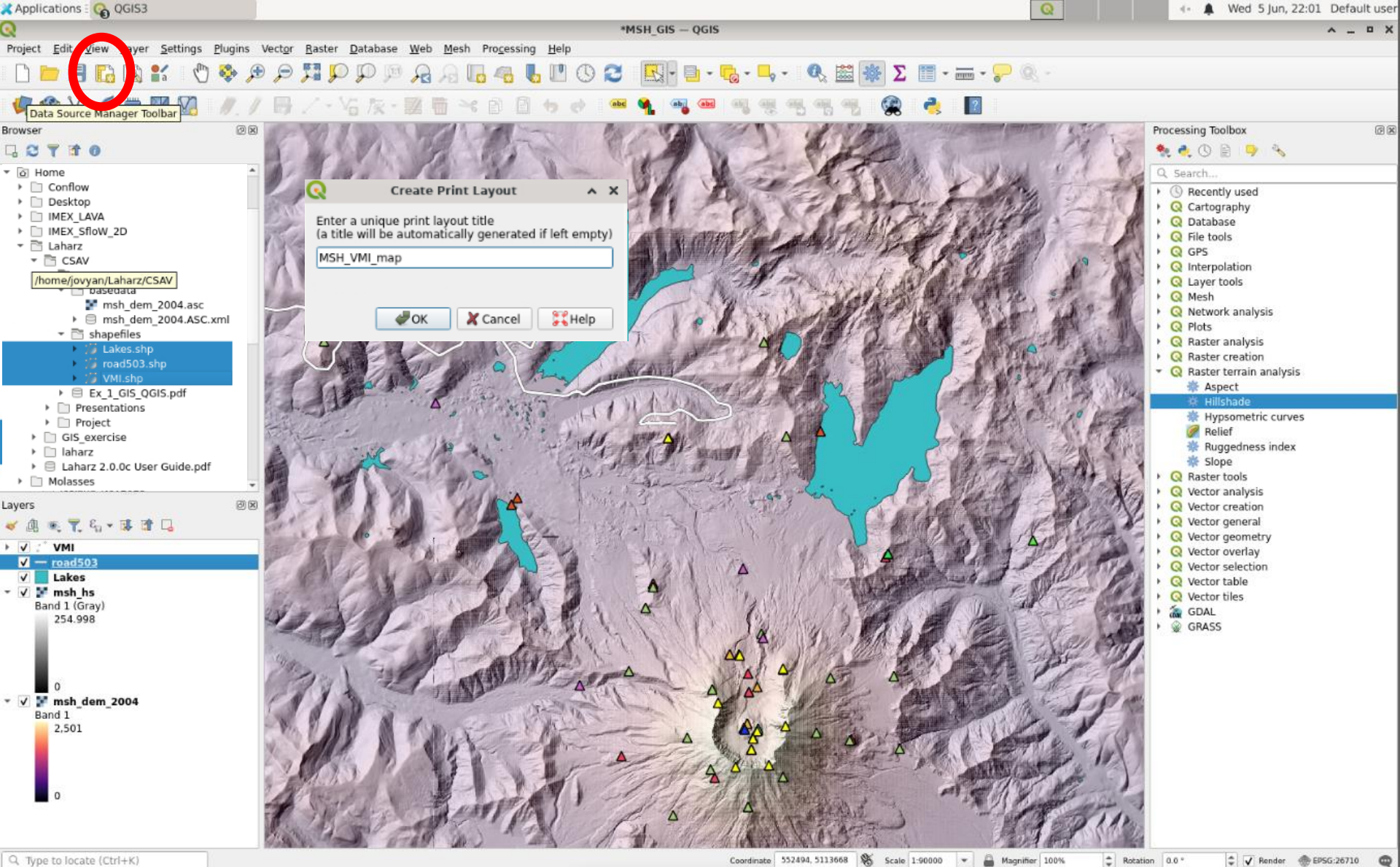


- Click the “Symbol” button to change the size and shape of the symbols. I chose 3 mm triangles
- Because station type is a categorical variable, choose “Random colors” for color.
- You can also change the colors for each symbol if you wanted
- Click “OK”
- Remember to save your project!

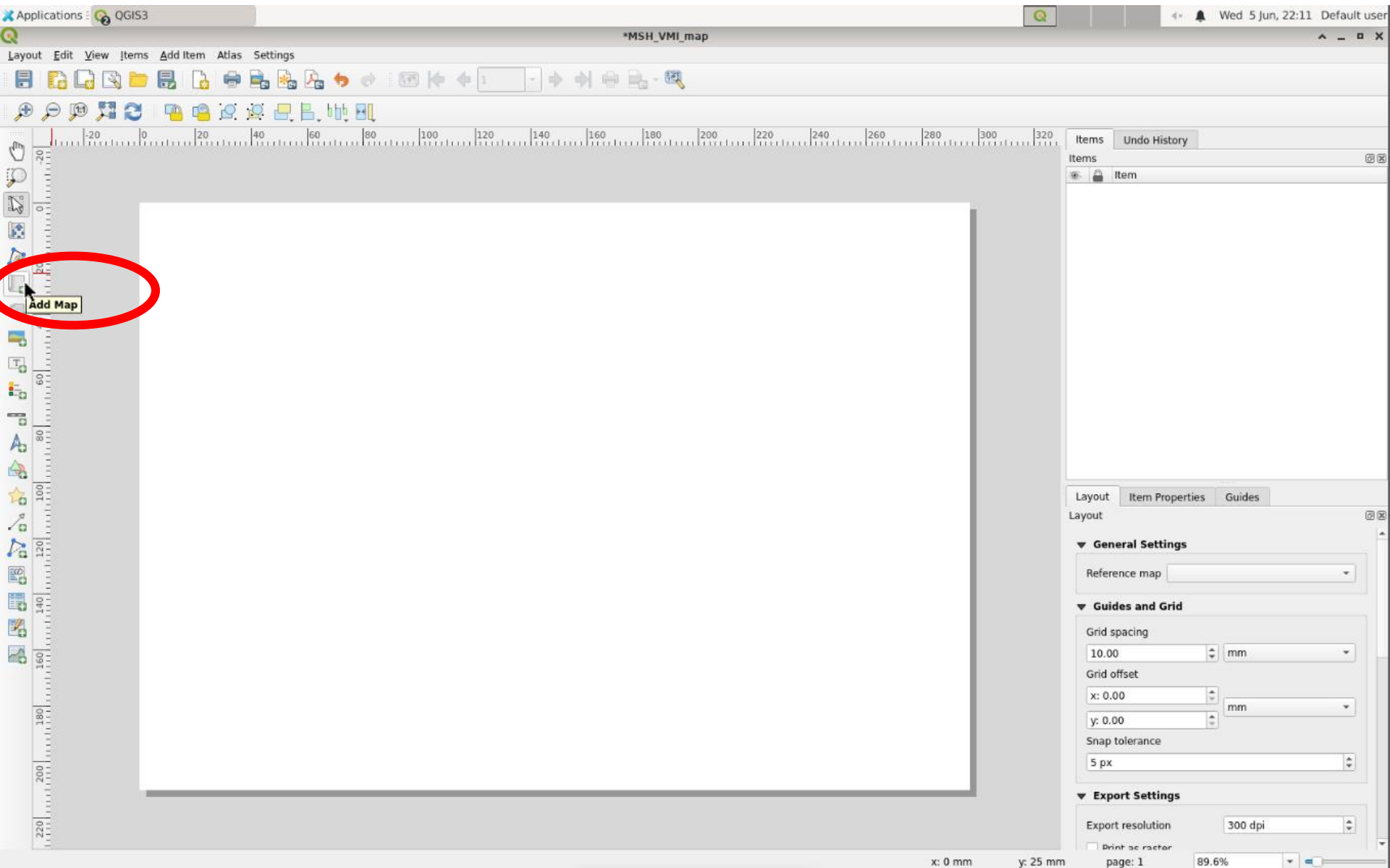


Exercise Goals

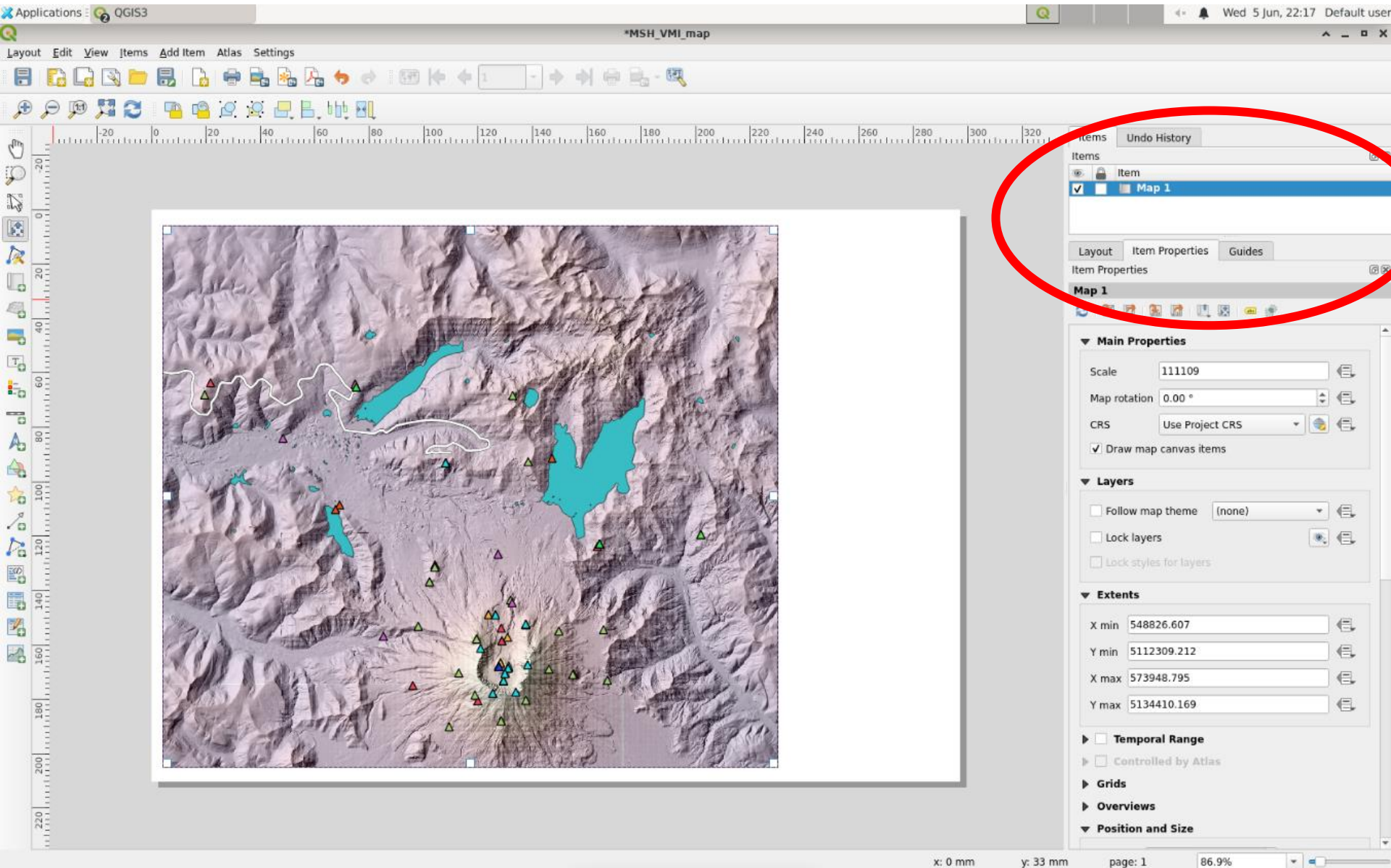
- Add Instrument data set and background information for context
- View Instrument attribute table
- Select instruments by a drawing rectangle
- Select instruments by their attributes
- Assign colors to instrument type
- **Add a legend to describe the meaning of colors**



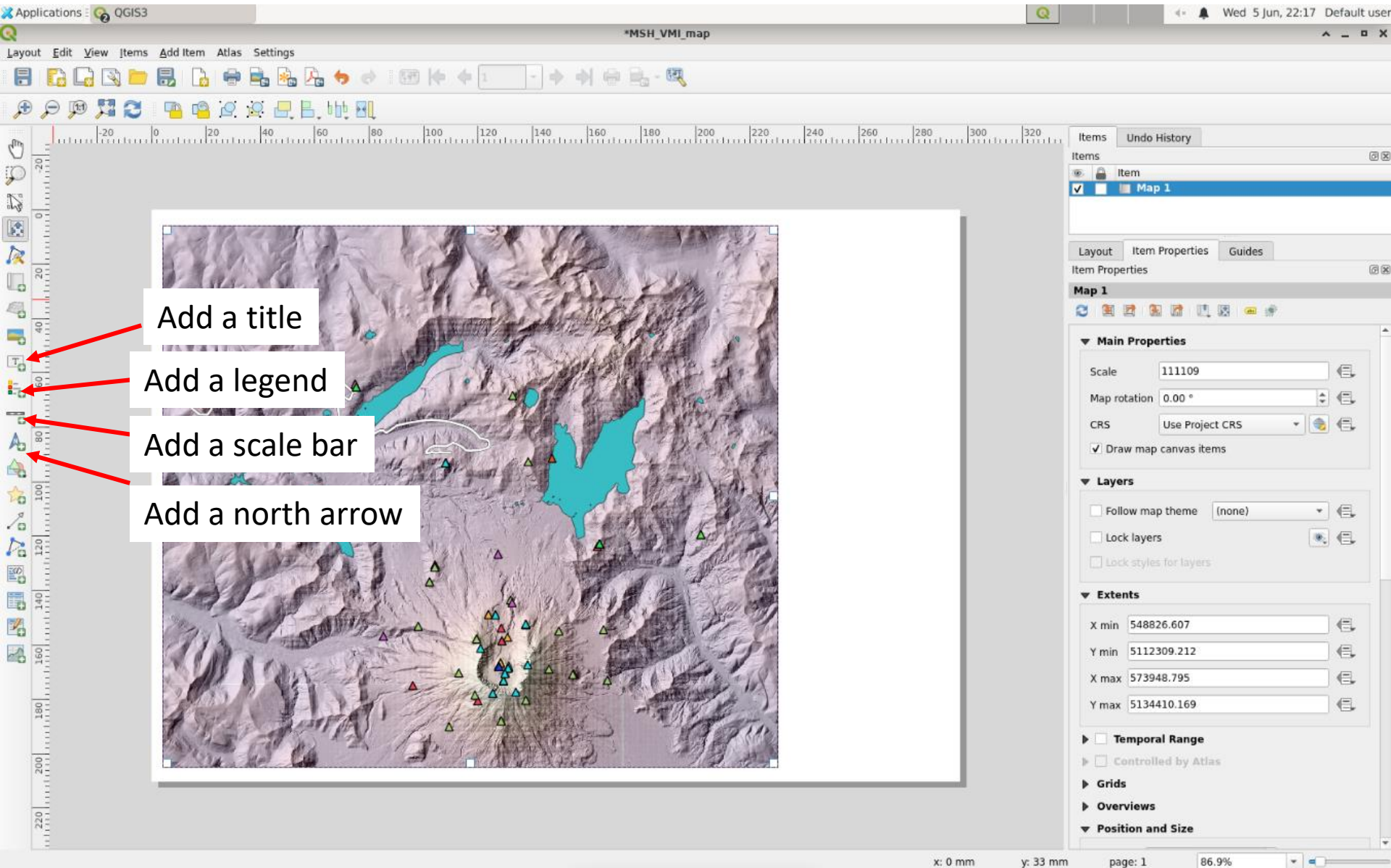
- Let's make a map!
- Click the "New Print Layout" button
- Enter a name for the new map. I chose "MSH_VMI_map"
- Click "OK"



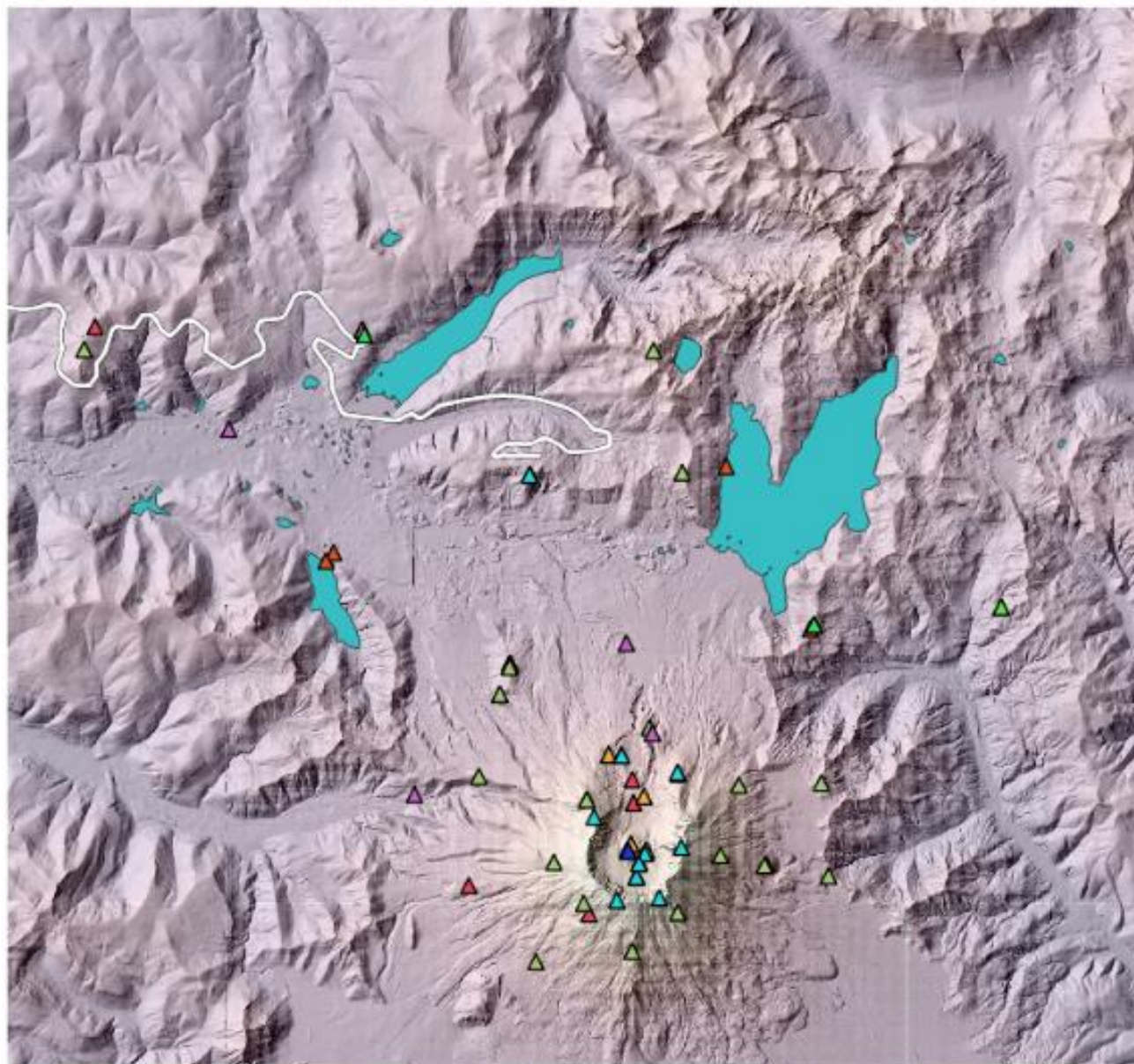
- First you need to add your map to the layout
- Click the “Add map” button
- Then draw a rectangle on the blank page to add the map



- You can adjust the scale and extent and other properties by selecting the map layer and “Item properties”



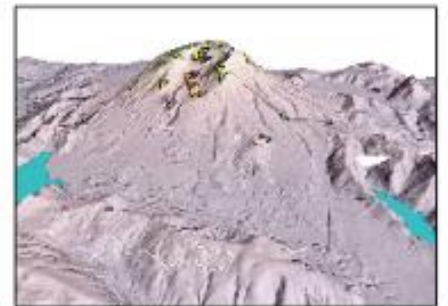
- You can add a legend for the stations and other layers
- Add a scale bar and/or coordinates
- Add a north arrow
- Play around with the options to make your map



Legend

Monitoring Instruments

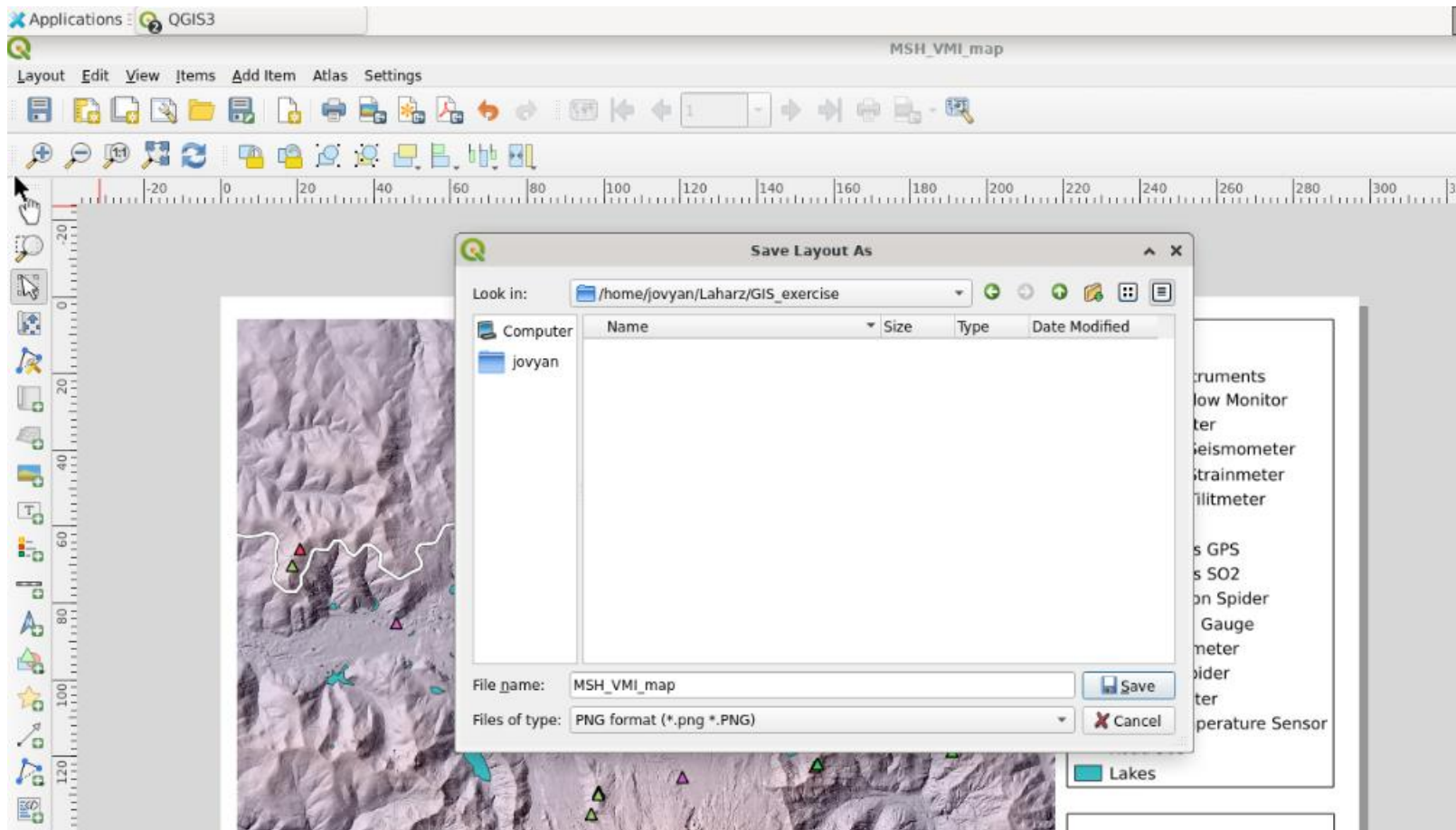
- ▲ Acoustic Flow Monitor
- ▲ Anemometer
- ▲ Borehole Seismometer
- ▲ Borehole Strainmeter
- ▲ Borehole Tiltmeter
- ▲ Camera
- ▲ Continuous GPS
- ▲ Continuous SO₂
- ▲ Deformation Spider
- ▲ Lake Level Gauge
- ▲ Microbarometer
- ▲ Seismic Spider
- ▲ Seismometer
- ▲ Water Temperature Sensor
- Road 503
- Lakes



0 2.5 5 km



- Save your final work!
- You can also export the map as an image



- Save your final work!
- You can also export the map as an image